

Predictability of U.S. Equity and Bond Excess Returns: a Point-in-Time Story

Stefan Bolta, FRM.

July 2026

Keywords: return predictability; point-in-time data; macro factors; forecasting; data snooping; risk premia
JEL Classification: G12, G14, G17, C53, C55, E44.

Abstract. Can macroeconomic data forecast monthly U.S. stock and Treasury-bond excess returns? Our attempt at an old problem is a leakage-free, point-in-time (PIT) panel of 506 macroeconomic series, on which we run a wide, pre-specified search across three forecast horizons, five bond maturities (and the S&P500), seven estimation windows and four sub-samples. We score every forecast against the naive historical-average benchmark. The result is a near-null, near-symmetric across targets: the macro data clear our bar in only 0.0% of equity trials and 5.0% of bond trials. The classic finance theory pools exactly zero, and the modest data-driven pool that shows potential is dismantled by the dependence-robust false-discovery control that a search this large demands. We identify a single evaluation cluster that passes the data-snooping test but fails to clear the dependence-robust false-discovery (FDR) control and does not persist beyond 2008.

1 Introduction and prior literature

The question of whether macroeconomic information forecasts aggregate asset returns out of sample is among the most contested in empirical finance. Welch and Goyal (2008) showed that the expanding-window historical mean is extraordinarily hard to beat OOS: of fourteen classical predictors, none deliver consistently positive OOS R^2 .¹ Campbell and Thompson (2008) partly rehabilitated predictability by imposing economic restrictions — most usefully a non-negativity truncation on the equity-premium forecast, worth 30–50 bp of OOS R^2 .² Rapach, Strauss and Zhou (2010) established that an equal-weighted combination of univariate forecasts dominates multivariate OLS, via a variance-reduction mechanism that is robust across samples and asset classes.³ Diebold and Shin (2019) survey the parallel shrinkage route — ridge, LASSO and elastic net — as the regularized alternative when predictor count is large relative to sample.⁴

¹Welch, I. and A. Goyal (2008). “A comprehensive look at the empirical performance of equity premium prediction.” *Review of Financial Studies* 21(4): 1455–1508.

²Campbell, J. Y. and S. B. Thompson (2008). “Predicting excess stock returns out of sample: can anything beat the historical average?” *Review of Financial Studies* 21(4): 1509–1531.

³Rapach, D. E., J. K. Strauss and G. Zhou (2010). “Out-of-sample equity premium prediction: combination forecasts and links to the real economy.” *Review of Financial Studies* 23(2): 821–862.

⁴Diebold, F. X. and M. Shin (2019). “Machine learning for regression: a tutorial with applications to financial market forecasting.” In *Forecasting* (Diebold).

On the bond side, the literature shows more optimism. Fama and French (1989) argued that the term spread and the default spread track business-condition variation in expected stock and bond returns.⁵ Cochrane and Piazzesi (2005) showed that a single tent-shaped linear combination of forward rates explains 30–44% of bond-return variance.⁶ Ludvigson and Ng (2009) demonstrated that macro principal components add a real-activity channel orthogonal to the forward-rate factor.⁷ Chen, Roll and Ross (1986) supply the canonical macro-risk pricing variables (industrial production, inflation, term and default premia) common to both targets.⁸

The contribution of this paper is practical and methodological at once: a leakage-free, point-in-time macro panel that lets any of these claims be tested honestly together with the discipline to keep a wide search from fooling us. We press the predictability question harder than the classic studies in three ways: a regime-adaptive set of short estimation windows, a high-dimensional all-variables predictor pool consumed by a shrinkage estimator class, and explicit multiple-testing control over the whole grid of results. Two further design choices support the evaluation: *each prediction target receives its own predictor pool*, and *acceptance is judged against the asset’s own buy-and-hold Sharpe rather than an arbitrary floor*. All data are public. Section 3 describes the data and Section 4 the findings.

2 Methodology

In plain terms the method does four things, in order. *First*, it builds an honest dataset: for every economic series we store not just its value but the date that value was first published, so we can always reconstruct exactly what was known on any past date. *Second*, it forecasts the future the way a real investor would have to — for each month in turn it fits a model on only the data available up to that month, predicts the next period’s return, then steps forward a month and repeats. *Third*, it asks whether each forecast actually beats a naive benchmark — simply assuming the future return equals the historical average — by enough to matter. *Fourth*, and this is what separates the exercise from a routine predictability scan: because we run thousands of these forecasts, we apply statistical controls so that a handful of lucky winners cannot pass themselves off as real discoveries.

Figure~1 is the roadmap. The rest of this section walks through each step, starting general and growing more technical as it goes: the point-in-time panel and the no-look-ahead guarantee (§2.2), how the data are transformed (§2.3) and which models are fit (§2.4), how each target’s predictor pool is built (§2.5), and the inference safeguards that decide what is believed (§2.6).

The equity excess return is the log S&P~500 total return over horizon h net of the 3-month bill; the bond excess return uses a duration approximation $r_{t,t+h}^{\text{bd}} = -D_n(y_{t+h}^n - y_t^n) + y_t^n h/12 - i_t^{3M} h/12$ on the maturity grid $n \in \{1, 2, 3, 5, 10\}$ with durations $\{0.97, 1.93, 2.85, 4.50, 8.00\}$. The estimators are given in model configuration (2.4). At each as-of date T the panel is rebuilt, transformed, fit on $[t_0, T - h]$, and used to forecast $T + h$; the loop advances one month. Sub-sample robustness uses four training windows (full, pre-2008, post-2008, post-2016) and seven estimation-window kinds: one expanding plus six rolling lengths $\{12, 24, 36, 48, 60, 84\}$ months (one to seven years). The rolling grid is deliberately front-dense and capped at seven years: predictability, if any, is plausibly a shorter-term, regime-dependent phenomenon. A trial is reportable only if it clears an

⁵Fama, E. F. and K. R. French (1989). “Business conditions and expected returns on stocks and bonds.” *Journal of Financial Economics* 25(1): 23–49.

⁶Cochrane, J. H. and M. Piazzesi (2005). “Bond risk premia.” *American Economic Review* 95(1): 138–160.

⁷Ludvigson, S. C. and S. Ng (2009). “Macro factors in bond risk premia.” *Review of Financial Studies* 22(12): 5027–5067.

⁸Chen, N.-F., R. Roll and S. A. Ross (1986). “Economic forces and the stock market.” *Journal of Business* 59(3): 383–403.

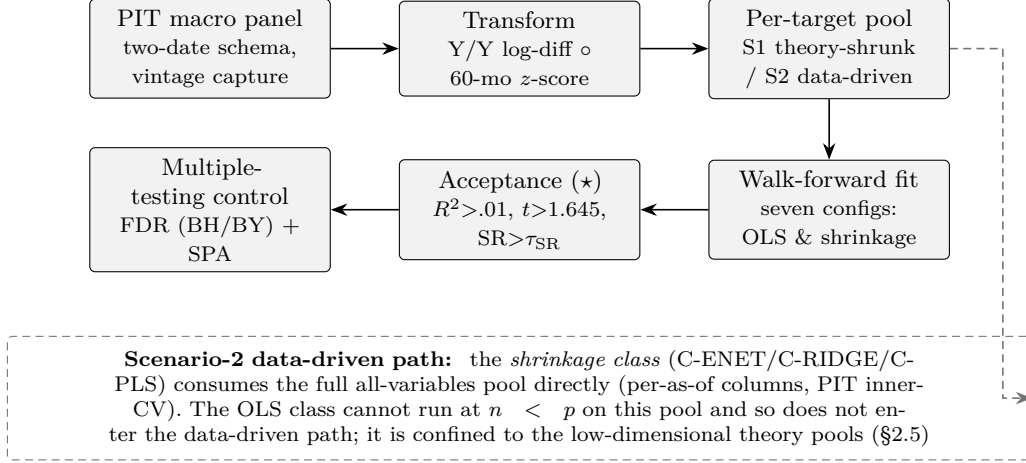


Figure 1: The data-processing and inference pipeline. Every forecast at date T uses only data released by T (the two-date PIT panel and the walk-forward jointly enforce this). The data-driven Scenario-2 pool is consumed directly by the shrinkage class (§2.5); every trial that clears the acceptance rule $(\star)$ is then put through the multiple-testing control of §2.6 before it is believed.

effective-sample floor $n_{\text{eff}} = n_{\text{train}}/h \geq 30^9$; the OLS class additionally requires $n_{\text{train}} \geq 4k$. A consequence, stated as a finding rather than hidden: at $h=3$ and $h=12$ the floor forces every short rolling window out, so regime-adaptive short-window prediction is feasible essentially only at $h=1$ — short-term predictability is a short-horizon phenomenon. Trials below the floor are excluded from every rate and disclosed separately. A *trial* is one point of this sweep — a (scenario, target, maturity, model configuration, horizon, estimation-window, sub-sample) tuple — and in total the grid yields 1,806 evaluable trials (276 equity, 1,530 bond). Where the experiment *begins* is set by the data rather than chosen: no forecast is scored until a pool’s youngest series has paid the full burn-in ladder — the year consumed by the year-over-year transform where it applies, the 60-month z -score window, then 120 months of minimum training — so the first scored forecast falls in December 2004 for the data-driven pools and between May 1991 and July 2016 across the theory pools. The last forecast was recorded on April 2026; Section 3 lays the timeline out in full.

2.1 Acceptance rule

A record is accepted for each set of predictors, and the null of no OOS predictability rejected, if three conditions are met in evaluation:

$$R_{\text{OOS}}^2 > 0.01, \quad t > 1.645, \quad \widehat{\text{Sharpe}} > \tau_{\text{SR}} \quad (\star)$$

The R^2 floor is calibrated to the 0.5–1.5% range that surviving Welch–Goyal predictors deliver; t is the Clark–West (2007) statistic for the nested comparison against the historical mean, and is the single t -value reported throughout the paper¹⁰. The third leg is a target-relative economic-significance bar: rather than an absolute floor, the strategy Sharpe must beat what the target itself delivers buy-and-hold, measured by the target’s own *PIT-expanding median Sharpe path*. For each as-of date τ in the test’s live OOS window $[t_{\text{eval}}, t_{\text{end}}]$, $\text{SR}(\tau) = (\bar{r}_{[t_0, \tau]} / \text{sd } r_{[t_0, \tau]}) \sqrt{12/h}$ is computed

⁹The target is h -overlapping, so the independent sample size is $\approx n_{\text{train}}/h$

¹⁰The t -value — whose one-sided 5% critical value is 1.645 — is standard-normal under its asymptotics, and is referred to a Student- t with $\lfloor n/h \rfloor - 1$ degrees of freedom once the long-horizon overlap of the equity refinements (§2.6) is accounted for.

on returns realized by τ , and the threshold is $\tau_{\text{SR}} = \text{median}\{\text{SR}(\tau)\}$, set per target (equity; each bond maturity)¹¹. The principled rule moves the headline by under one percentage point on either target (Section 4), so the near-null is robust to the acceptance criterion — it is a property of the data, not of the threshold.

2.2 The point-in-time panel and the no-look-ahead guarantee

The panel is this paper’s main contribution, so we describe it before the models that consume it. Its design principle is one sentence: **store every observation together with the date it was first published, never overwrite it, and let the accessor reconstruct the world as it looked on any past date.** Everything below follows.

The two-date schema. Every stored row carries two dates, not one: a `reference_date` (the period the number describes) and a `release_date` (the day that number first became public), alongside the `value`, a `vintage_id` identifying which revision it is, the authoritative `source`, and an `ingest_timestamp`. A single invariant holds on every row and is checked at ingestion — `release_date` $\geq$ `reference_date`: a figure cannot be known before the period it measures has even ended. That second date is the cheap column everything else rests on, and it cannot be recovered after the fact, which is why it is captured at ingestion rather than bolted on later.

The accessor and PIT invariance. Let $\mathcal{F}_T$ be the information set of every observation whose `release_date` is on or before calendar date T . The panel accessor returns a wide matrix $\mathbb{X}^{(T)} \in \mathbb{R}^{M \times N}$ whose cell (t_m, n) holds the latest vintage of factor n for reference month t_m that had been *released* by T — a strict per-cell projection onto $\mathcal{F}_T$. The construction guarantees PIT invariance: a record computed as-of T_1 is identical when the panel is recomputed at any later $T_2 > T_1$, so no revision leaks backward into a training window and re-running the walk-forward years later reproduces every historical forecast exactly. The panel ships with a single canonical accessor, `get_pit_value(factor_id, as_of_date)`, that returns any series exactly as it stood on a given date — the matrix $\mathbb{X}^{(T)}$ is that same query applied across all series at T — so downstream factor-model work calls one function and inherits the no-look-ahead guarantee for free.

The two leakage modes it closes. PIT invariance shuts the two ways a macro backtest silently inflates itself. *Vintage leakage* uses a revised figure before its revision was public — training on today’s heavily-revised 2008 GDP, say, which nobody had in 2008. *Release-timing leakage* uses a figure before it was released — assuming January CPI is in hand on 31 January when it actually prints in mid-February. Both are prevented architecturally rather than patched after the fact: a model fit at T sees only rows with `release_date` $\leq T$, and for those the vintage that was current at T . Either leak, left open, raises backtest Sharpe ratios for nothing.

Vintage capture where it is free. For series whose revision history is publicly archived we ingest every vintage, not just the latest: FRED’s ALFRED archive for revised macro aggregates, the Philadelphia Fed Real-Time Data Set for the SPF and key activity series, and the BEA archive for the national accounts. Where no vintage archive exists we capture the authoritative release calendar instead — enough to close release-timing leakage even where vintage leakage cannot be reconstructed. Which guarantee a series enjoys is recorded as its quality tier (below), so the strength of every backtest is visible at the level of each input rather than assumed for the panel as a whole.

¹¹We use a target-relative bar rather than an absolute $\widehat{\text{Sharpe}} > 0.5$ floor: a fixed bar rewards a test for merely riding a bull market, whereas a strategy that adds value must beat holding the asset. Equity’s median path runs ≈ 0.47 (just under the old floor, so the bar relaxes slightly); the bond maturities straddle 0.5 (0.43–0.74), the long end posing a harder bar because the 1990–2020 bond bull market lifted realized Sharpes.

2.3 Data transformation

Write $\mathbf{x}_t$ for the transformed predictor vector at month t . Two transform classes are applied before any estimator sees the data.

1. *Flow / index*: Each series expressed in nominal terms (industrial production, the price indices, payrolls, etc.) is transformed to year-over-year log-difference form in order to remove the unit root and the seasonal. *Level / spread* series (yields, the term and default spreads, the volatility and financial-conditions composites, valuation ratios) skip this step.

$$\Delta_{12} \log x_t = \log x_t - \log x_{t-12} \quad (1)$$

2. *Standardization*: Every series is then standardized by a trailing 60-month rolling z -score, computed strictly in-sample at each as-of date so the standardization is itself PIT.

$$z_t = (x_t - \hat{\mu}_{t-59:t}) / \hat{\sigma}_{t-59:t} \quad (2)$$

3. *Targeted winsorization*: No general winsorization is applied to the predictor panel — the z -score above is left untruncated ¹².

2.4 Model configurations

Seven configurations span the surveyed literature, in two classes. An OLS class (C-OLS, C-COMB, C-CT, C-CP) requires a pre-reduced predictor set and runs only where $n > p$. A shrinkage/supervised-reduction class (C-ENET, C-RIDGE, C-PLS) is built for the high-dimensional regime, consumes the wide all-variables pool directly, and self-regularises through a strict point-in-time hyperparameter search (below) so that complexity cannot manufacture predictability. Throughout, $r_{t,t+h}$ is the target return over horizon h and $\mathbf{x}_t$ the predictor vector at month t .

2.4.1. C-OLS — Welch–Goyal (2008) baseline.

$$r_{t,t+h} = \alpha + \beta' \mathbf{x}_t + \varepsilon_{t,t+h} \quad (3)$$

Ordinary least squares, but inference uses Newey–West (1987) heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors at bandwidth $L = \lfloor 1.5h \rfloor$. The choice is load-bearing at $h > 1$: overlapping multi-period returns induce an $\text{MA}(h-1)$ autocorrelation in the residuals that plain OLS errors ignore and that would otherwise inflate the t -values. HAC widens the standard errors — a more conservative test — while leaving the point forecast unchanged.

2.4.2. C-COMB — Rapach, Strauss & Zhou (2010).

$$\hat{r}_{t,t+h} = K^{-1} \sum_{k=1}^K \hat{r}_{t,t+h}^{(k)}, \quad \hat{r}_{t,t+h}^{(k)} = \hat{\alpha}_k + \hat{\beta}_k' \mathbf{x}_{k,t} \quad (4)$$

¹²The only value-clipping in the pipeline is a structural-break clip on two of the SPF surprise inputs (Section 2.7): observations with $|\text{CPI surprise}| > 50$ (the 1988 BLS base-year rebase, 1967=100 $\rightarrow$ 1982–84=100) or $|\text{INDPRO surprise}| > 15$ (the 2002 BLS methodology revision) are set to missing rather than allowed to enter as signal.

The equal-weighted mean of K single-regressor OLS forecasts. Pooling is a variance-reduction device: by never estimating the predictors jointly it sidesteps the multivariate over-fit that sinks C-OLS in wide pools, and it is robust across samples and asset classes.

2.4.3. C-ENET — elastic net (Zou & Hastie 2005; Diebold & Shin 2019).

$$\hat{\beta} = \arg \min_{\beta} \sum_t (r_{t,t+h} - \alpha - \beta' \mathbf{x}_t)^2 + \lambda \left[\frac{1-\rho}{2} \|\beta\|_2^2 + \rho \|\beta\|_1 \right] \quad (5)$$

Mixing $\rho = 0.5$; the penalty λ is chosen by the PIT-expanding inner cross-validation described below (the λ grid taken from the `glmnet` path on the outer-training window, so no look-ahead). The ℓ_1 term self-selects from the full pool, so C-ENET is the regularized counterpart to the explicit Section 2.5 selector — and a check on it.

2.4.4. C-CT — Campbell–Thompson (2008), equity only.

$$\hat{r}_{t,t+h} \longrightarrow \max(0, \hat{r}_{t,t+h}) \quad (6)$$

The C-OLS forecast under a positivity truncation, imposing the economic restriction that the equity risk premium is non-negative; in the original study the restriction is worth 30–50 bp of OOS R^2 .

2.4.5. C-CP — Cochrane–Piazzesi (2005), bond only.

$$\text{CP}_t = \gamma' \mathbf{f}_t, \quad r_{t,t+h}^{\text{bd}} = \alpha + \beta \text{CP}_t + \varepsilon_{t,t+h} \quad (7)$$

A single forecasting factor formed as the fitted linear combination of forward rates $\mathbf{f}_t$, collapsing the whole yield curve into one return-predictive index.

2.4.6. C-RIDGE — ridge regression (Hoerl & Kennard 1970; the shrinkage route of Diebold & Shin 2019).

$$\hat{\beta}(\lambda) = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1} \mathbf{X}'\mathbf{r}, \quad \lambda \in \{10^{-3}, \dots, 10^3\} \text{ (15-pt log grid)} \quad (8)$$

The intercept is un-penalised and predictors are standardized within each fit. Ridge keeps all p predictors and is the high-dimensional workhorse for the wide pool: under the inner CV it *self-regularises* — when the effective sample is small, any non-trivial fit validates poorly and the CV selects heavy shrinkage (large λ), so the forecast collapses toward the training mean and no false signal is manufactured.

2.4.7. C-PLS — partial least squares (Wold 1975; Kelly & Pruitt 2013).

$$\mathbf{w}_k = \arg \max_{\|\mathbf{w}\|=1} \text{Cov}(\mathbf{X}_k \mathbf{w}, \mathbf{r}), \quad \hat{r}_{t,t+h} = \hat{\alpha} + \sum_{k=1}^m \hat{\gamma}_k t_{k,t}, \quad t_k = \mathbf{X}_k \mathbf{w}_k \quad (9)$$

Supervised dimension reduction: PLS extracts the few directions of the predictor space most covarying with the target (deflating $\mathbf{X}$ after each component), then regresses on the first m scores. Where PCA compresses by predictor variance alone, PLS compresses *toward* the target¹³; the number of components $m \in \{1, 2, 3\}$ is capped low and chosen by the inner CV. Being supervised, it is the most prone to overfit, hence the ≤ 3 -component cap.

Hyperparameter selection (PIT inner CV). For the shrinkage class, λ (ridge / elastic net) and m (PLS) are chosen by a strict point-in-time expanding inner cross-validation *within* each outer training

¹³Ludvigson–Ng 2009 use the PCA variant for bond premia.

window: validation points step forward h -spaced, each inner fold trains only on rows preceding its validated point by at least h , and the value minimizing aggregate validation loss is refit on the full window before forecasting $T + h$. No future information enters the choice — this is the guardrail that keeps the high-dimensional class honest. For the unbalanced pool, each shrinkage fit at T uses the per-as-of available columns (non-NA at x_T and complete over the training window); the dropped-column count is surfaced, never silently absorbed.

The OLS class (C-OLS, C-COMB, C-CT) cannot run at $n < p$ on the all-variables pool, so it does not figure in the data-driven results; it is confined to the low-dimensional theory pools of Scenario 1. The shrinkage class (C-ENET, C-RIDGE, C-PLS) consumes the full all-variables pool directly, selecting internally via its penalty or supervised reduction, and is the primary — and only — data-driven path.

2.5 Target Scenarios

For each target we run two scenarios, one anchored in the literature and one data-driven.

2.5.1 Scenario 1 — Academic theory

A small pool lifted directly from the relevant canon; each row of the table below is a published forecasting design, reproduced on the PIT panel.

Table 1: Scenario 1 — the academic-theory pools replicated, by target. Each row reproduces a published forecasting design on the PIT panel.

Target	Authors (year)	Procedure	Variables
Bond	Fama–French (1989)	C-OLS, C-COMB	term spread (10Y–3M), default spread (BAA–AAA)
Bond	Cochrane–Piazzesi (2005)	C-CP	five forward rates (1–5Y implied forwards)
Equity	Welch–Goyal (2008) $\cup$ Chen–Roll–Ross (1986)	C-OLS, C-COMB, C-ENET, C-CT	inflation (CPI), industrial production, consumption deflator (PCE); 3M and 10Y Treasury yields, term spread, default spread, equity-return variance (VIX), 10Y expected inflation

2.5.2 Scenario 2 — Data first approach

Here we assume no prior knowledge of any theory: we ingest available data and let the estimator decide. The pool holds **114 candidate features** (83 macro series + 18 SPF nowcast levels + 13 practitioner constructs); Section 3 places these counts in the full series-to-feature ladder. The reason the equity target must be a month-end total return rather than a within-month average — and the artifact that follows from getting it wrong — is documented where it bites, in Part I.

Because the pool is wide and the histories are unbalanced (series start at different dates), it is consumed by the shrinkage/supervised class (C-ENET, C-RIDGE, C-PLS), which takes the full 114-feature pool directly, seeing at each T only the per-as-of available columns; no imputation is applied — the estimator handles the unbalanced panel. The OLS class (C-OLS, C-COMB, C-CT) cannot run at $n < p$ on a pool this wide, so the data-driven evaluation rests entirely on the shrinkage class; the OLS configurations enter only the low-dimensional theory pools of Scenario 1.

A component-regression route to the same pool — compress the collinear block to a few principal components, then regress — is the classical alternative, and the one that historically found macro-to-bond predictability (Stock & Watson 2002; Ludvigson & Ng 2009). We do not run it as a separate method, because on this pool it offers nothing the shrinkage class does not already provide: a supervised component regression maximizes covariance, which is exactly partial least squares (Frank & Friedman 1993), so it is subsumed by the C-PLS configuration. Where faint short-end bond predictability appears, it is captured by these smooth shrinkers rather than by any selection rule (Part II); the supplementary methods note records the component-selection variants we examined and set aside.

2.6 Inference safeguards: leakage control and multiple-testing correction

Two distinct ways to manufacture a false positive must be closed. The first is *leakage*, handled architecturally and described in Section 2.2 so neither *vintage leakage* nor *release-timing leakage* can inflate a result. The two channels would not bias a backtest the same way, and we quantify each. The *release-timing* channel is the within-month-average equity artifact that manufactures acceptances outright (see *footnote 24*). The *vintage* channel is isolated in Annex A.2¹⁴. Release timing changes *which* trials pass; vintage revision only exaggerates the *magnitude* of those already passing — and the two-date schema closes both. The second is *data snooping*: the evaluation sweeps a large grid of trials — scenario $\times$ target $\times$ maturity $\times$ config $\times$ FS iteration $\times$ horizon $\times$ estimation window $\times$ sub-sample — and the trials are heavily correlated (overlapping long-horizon returns; nested windows). A single $t > 1.645$ is then not evidence: with a family of J correlated trials, some cross the bar by chance. We therefore subject any surviving acceptance to explicit multiple-testing control before believing it¹⁵.

Two complementary questions are then put to anything that clears the acceptance bar, and each maps to exactly one procedure. The first — *how many of the winners are real, not luck?* — is a false-discovery question, answered by false-discovery-rate (FDR) control. We treat FDR as a single control read at two stringency levels: the Benjamini–Hochberg (1995) bound, valid under positive dependence, and its Benjamini–Yekutieli (2001) tightening, valid under arbitrary dependence¹⁶. Because our trials are strongly correlated, the dependence-robust Benjamini–Yekutieli level is the headline and Benjamini–Hochberg the more lenient comparison — two readings of the one FDR control, not two separate stages. The second — *is even the single best model real, once we account for how many we tried?* — is answered by the Hansen (2005) superior-predictive-ability (SPA) test, which compares the best model against a bootstrap distribution built over the entire search. The two are not redundant rather complementary: FDR polices the breadth of a discovery, SPA the survival of its strongest point.

The SPA test needs the pointwise forecast paths. The equity Scenario-2 trials keep theirs, and we retained the paths for all five bond maturities, so it runs across the full curve on both targets. The equity paths support two further refinements that use the equity one-period innovation: an out-of-sample accuracy test corrected for the overlap that long-horizon returns induce, and an in-sample check of whether the forecast actually tracks the realized return.¹⁷ A genuine signal

¹⁴Re-running the data-driven shrinkage class on a counterfactual panel that substitutes final-revision values while holding release timing fixed inflates every fit statistic — the surviving pre-2008 short-end bond signal’s best tradable Sharpe rises from 0.82 to 1.19 — yet leaves the accept/reject map unchanged (0 of 18 trials flip).

¹⁵The machinery is applied where a positive remnant remains, in Part II: the theory-pool and surprise results are negative, so correction only deepens them.

¹⁶Both cap the expected fraction of false positives among the accepted trials. For J ordered p -values $p_{(1)} \leq \dots \leq p_{(J)}$ at level q , BH rejects the largest i with $p_{(i)} \leq (i/J)q$; BY tightens the bound to $p_{(i)} \leq (i/J)q / \sum_{l=1}^J l^{-1}$.

¹⁷The overlap correction refers the t -value to a Student- t on $\lfloor n/h \rfloor$ effective observations; the in-sample check

should clear all of these; a fluke should not.

2.7 Surprise constructs

The surprise layer adds four macro surprises — realized minus SPF consensus nowcast — for real GDP, CPI, unemployment, and industrial production. The realized leg uses the first vintage (the value the market saw at release); the forecast leg is the SPF mean nowcast; the surprise inherits the release date of the realized leg, so the construct is itself PIT-correct. These are spliced into the equity theory pool and into Scenario 2; a leave-one-out (LOO) ablation isolates each surprise’s marginal contribution.

3 Data

The PIT store holds **506 ingested series**. A *series* (equivalently, a *variable*) is one raw data series; transforming it — a year-over-year log-difference, a rolling z -score — does not make it a new variable. A *feature* is a column actually handed to an estimator: either a transformed series, or a construct *derived* from several series. Keeping series and features apart is what lets the counts below reconcile to a single total. Sources are exclusively public-tier: FRED/ALFRED, the Philadelphia Fed Real-Time Data Set and Survey of Professional Forecasters, CBOE volatility indices, the Baker–Bloom–Davis policy-uncertainty series, Federal Reserve regional and financial-conditions composites, and the Shiller/AAIL/Wurgler academic pages. Of the 506 series, *308* are SPF survey nowcasts (the FORECASTS family) and the remaining *198* are non-forecast series. Quality tier and native frequency are presented below, the complete feature inventory is in Annex A.3.

Table 2: Corpus by PIT quality tier (series type).

Series type	Count
Total series	506
VINTAGE	440
RELEASE-DATED	66

Table 3: Corpus by native frequency.

Frequency	Count
Quarterly	387
Monthly	91
Weekly	8
Daily	20

The series-to-feature ladder. Three counts matter, at three different levels, and each is a clean partition. *(i) Concept.* The project’s literature compilation curates **157 research factors** in an 18-family economic taxonomy — the conceptual inventory the data are meant to instrument. *(ii) Series.* The store holds **506 ingested series** = 308 SPF FORECASTS + 198 non-forecast series. Of these, **101 are used directly** and **405 are held out** — 290 SPF series beyond the 18 mean nowcast levels we admit, plus 115 raw real-time-vintage (RTDSM) stacks whose as-of- T value the panel already carries. *(iii) Feature.* **119 features actually enter a model**, across all pools = the

is an efficiency regression of the realized return on the forecast (Mincer & Zarnowitz 1969) with Hodrick (1992) overlap-robust standard errors. These two legs use the equity one-period innovation and are reported for equity; the bond test is the SPA on the retained bond paths, run per maturity.

101 series used directly + **18 derived features** (the 13 practitioner constructs¹⁸, the two yield spreads, and the three SPF surprises). There is no feature-expansion step — each series maps to exactly one feature, with no lags or interactions — so the pipeline only ever *narrows*: 506 series → 119 features. The largest single pool is the data-driven Scenario-2 pool of **114 features** (83 macro + 18 SPF M0 + 13 practitioner)¹⁹.

Quality tiers: what each series is good for. A point-in-time backtest is only as honest as the provenance of its weakest input, so every series carries a quality tier stating which leakage guarantee it enjoys. A **VINTAGE** series carries its full revision history and supports the strictest reconstruction: any past as-of view is exact, so both leakage modes are closed.²⁰ A **RELEASE-DATED** series carries an authoritative release calendar but only latest-revision values; it closes release-timing leakage but not vintage leakage, and is suitable for most practical work with that caveat made explicit. A **REFERENCE-DATED-ONLY** series, where even the release date is estimated, would be the weakest tier, flagged for cautious use or exclusion. On the present public-tier corpus the realized split is 440 VINTAGE and 66 RELEASE-DATED, with 0 reference-dated-only: the proprietary practitioner indices that would fall in the weakest tier are deferred to a later vendor-access phase and play no part in the results here.

The store is audited, not trusted. An automated audit re-checks the panel on every one of the 506 series rather than taking the schema on faith: the `release_date ≥ reference_date` invariant on every row, the plausibility of each series’ realized release lag against its documented calendar, and the monotonic consistency of successive vintages. The monotonicity and vintage-presence checks pass on every series to which they apply²¹ while the remaining series are by-design exemptions — survey nowcasts have no prior vintage to compare, and release-dated series carry no vintage history. The release-lag check flags 102 series: 93 are the held-out real-time vintage stacks²² and the remaining 9 are FRED series whose flags fall on documented government-shutdown release delays. This audit is what secures the downstream numbers as honest out-of-sample results: the guarantee is enforced and verified at the data layer, before any model is fit.

The sample timeline, once. Because the panel is unbalanced, when the experiment starts is not one date but one *rule*, applied per pool: the first *scored* forecast falls where the pool’s youngest series has been available long enough to pay the full burn-in ladder — twelve months consumed by the year-over-year transform (flow series), sixty by the rolling z -score ($W_Z = 60$), and 120 further months of minimum training ($t_{\text{eval}} = t_{\text{start}} + 120$). Availability here means *point-in-time* availability: a series enters the panel at its first archived release, not its first reference date, so a long back-history published late buys no head start. Concretely, the data-driven Scenario-2 cells

¹⁸The practitioner constructs operationalize T. Rowe Price’s published (2020) tactical asset-allocation dashboard — a panel of macro signals (equity volatility, core CPI, unemployment, the real 10-year yield, oil, the trade-weighted dollar, investment-grade and high-yield credit spreads, gold, euro/yen/EM-currency crosses, global industrial production, a manufacturing-PMI reading and euro-area core inflation) together with a handful of relative-value crosses — rebuilt entirely on public-tier data. Where the dashboard’s exact instrument is a private vendor series, we substitute a documented public proxy. They enter evaluation only as derived features in the data-driven Scenario-2 pool. One important note, dashboard was built to forecast *pair-relative* returns rather than the absolute risk premium.

¹⁹Annex Table A.3 enumerates, audits and sources every one of the 119 features and is the single source of truth.

²⁰That history is not a formality. Across the 130 genuinely-revised series (ALFRED and RTDSM; survey rounds excluded), first releases are materially restated — the median series has a revision-to-signal ratio of 0.29 and 25% exceed 0.5 — and the revisions are two-sided (53% upward), so they cannot be undone by assuming first prints are biased; only the vintage record recovers them. The familiar case is 2008-Q4 real GDP, first reported at −3.8% annualized and now −8.5%. Annex A.2 audits every series, measuring revisions on the year-over-year growth the model consumes (Croushore and Stark 2001; Aruoba 2008).

²¹Pass rate: 198/198 and 440/440.

²²These belong to 1960s–80s release conventions and were never used as model features.

anchor the clock on the deep cohort — the series already reporting by January 1990 — which puts their first scored forecast at December 2004; younger features join the unbalanced pool per as-of date as they become available (§2.5). Each theory pool starts where its own youngest input allows: May 1991 for Cochrane–Piazzesi (bound by the 2-year yield, whose history begins in June 1976), August 1996 for Fama–French (the 3-month constant-maturity yield, September 1981), and only July 2016 for the Welch–Goyal $\cup$ CRR equity pool (the PCE deflator, whose real-time archive opens in 2000) — a start so late the equity theory pool has no pre-2008 out-of-sample window at all. Every out-of-sample window then runs to the 30 April 2026 as-of date, split at January 2008 and January 2016 for the sub-samples and shortened per horizon by the effective-sample floor — which, at $h = 12$, removes the short rolling windows entirely.

Feature versus trial. These 119 features generate a far larger count of statistical tests: the full sweep is **6,768 trials** (a trial being one feature-set $\times$ config $\times$ window $\times$ horizon $\times$ sub-sample combination), of which **1,806** are evaluable after the feasibility floor. The multiple-testing universe of Part II (the FDR and SPA tests) is over *trials*, not features — the distinction the false-discovery correction is built to police.

4 Results

4.1 Part I — The forecasts

4.1.1 The apparent signal, before testing

Table 4: Acceptance by scenario and target, level pools (eqn $\star$).

Scenario	Trials	Accept	Pass %
S1 / bond / Cochrane-Piazzesi	135	0	0.0
S1 / bond / Fama-French	270	0	0.0
S1 / equity / WG+CRR	66	0	0.0
S2 / bond (data-driven)	360	38	10.6
S2 / equity (data-driven)	72	0	0.0
All	903	38	4.2

Three findings stand out. **First, the theory canons are dead on the modern point-in-time sample.** The Welch–Goyal $\cup$ Chen–Roll–Ross equity pool clears the bar ($\star$) in 0 of 66 trials; the Fama–French (1989) term-and-default-spread bond pool in 0 of 270, across every maturity, config, horizon, window and sub-sample; the Cochrane–Piazzesi forward-rate factor in 0 of 135. The most-cited macro-return prediction claims in the literature simply do not replicate once each forecast is restricted to data known at the time. Even the single strongest trial of each canon — the best it manages anywhere in the sweep — is rejected by the acceptance rule (Annex Figure~5): the equity pools barely move the forecast, and the bond pools’ apparent fit is an isolated window-and-regime trial rather than a durable signal.

Second, the data-driven pool does modestly better than theory on bonds, and no better on equity — and even the bond edge is small and measured before any correction for the size of the search: 0.0% of equity trials clear the bar (the same flat zero as the theory pools) and 10.6% of bond trials. That bond acceptance comes from combining many predictors on the wide pool, not from any one of them; Part II then asks whether it survives multiple-testing control, and as breadth it does not — with one regime-confined pre-2008 short-end exception that clears the SPA but not the false-discovery control.

Third, what bond predictability there is sits at the short end — it falls off steadily with maturity, concentrated in the one-year bond and fading to zero at the long end (the *bond acceptance rate by maturity* panel of Annex Figure~4). the ten-year bond accepts in 0.0% of its evaluable trials and the five-year in 1.3%, while the one-year reaches 15.4% — the modest bond signal concentrates at the short end. Two horizon patterns matter for what follows, and both are short-horizon. On the bond side the monthly ($h=1$) Scenario-2 rate (12.3%) is the strongest, against 1.7% at $h=3$, with $h=12$ infeasible under the effective-sample floor²³ — the bond signal is a short-window, short-maturity, monthly-horizon phenomenon. On the equity side, Scenario-2 acceptance is 0.0% at $h=3$ and 0.0% at $h=1$ — a flat zero at every feasible horizon, with $h=12$ infeasible under the same floor. That the equity column is empty is itself a result: a naïve within-month-average equity target manufactures a large $h=1$ signal that is purely an averaging artifact.²⁴ The shrinkage class regularises itself away on the wide equity pool rather than manufacturing signal — the elastic net, for instance, accepts zero equity trials with negative mean OOS R^2 .

Before subjecting any of this to testing, we add one more layer to the apparent signal: the macro surprises. Adding the four surprises leaves data-driven acceptance essentially unchanged. The shrinkage estimators absorb the three surprise columns into the roughly 114-feature pool with no material effect — the expected behaviour of a regularized fit on a wide pool, and one more reason the data-driven path stands on that class. Whether any of the remaining apparent signal is a real discovery is precisely the question Part II’s multiple-testing control settles.

Table 5: Surprise augmentation on the shrinkage class: accepted trials, surprises minus levels (acceptance rule $\star$). The increment is essentially zero — once consumed by the regularized fit on the wide pool, the macro surprises add no acceptance.

Scenario	Trials	Levels	+Surprises	Delta
S1 / bond / CP	135	0	0	0
S1 / bond / FF1989	270	0	0	0
S1 / equity	66	0	0	0
S2 / bond	360	38	38	0
S2 / equity	72	0	0	0
All	903	38	38	0

4.1.2 What each procedure leans on

A natural worry is that the data-driven acceptance rests on one or two lucky variables. It does not. The shrinkage estimators that carry the data-driven path — ridge, partial least squares, and the elastic net — impose no selection at all: they consume the full 114-variable pool through their own penalties and spread their weight so thinly that no single variable reaches 3% of any fit. There is no dominant predictor to point to; the apparent signal, where it appears at all, is an emergent

²³Which requires $n_{\text{train}}/h \geq 30$, so every short rolling window is dropped at the annual horizon.

²⁴Earlier versions of this exercise used the Shiller series’ within-month *average* price as the equity target. A calendar-month average overlaps the daily prices of neighbouring months, so it is mechanically correlated with lagged predictors and manufactures spurious short-horizon “predictability” that is strongest at $h=1$. Compounded by a candidate pool that silently auto-included the index’s own averaged level (a circularity), this produced an apparent 52.6% equity-acceptance headline — including an elastic-net $h=1$ strategy reporting a Sharpe of ≈ 2.4 and a “cluster of ten” standalone sentiment predictors led by `AAII_BULLBEAR` and `VIXCLS`. Replacing the target with the month-end total-return index (`SP500TR`) and purging the index’s own level from the candidate pool collapses all of it (equity acceptance 52.6% $\rightarrow$ 0.0%). The lesson generalizes beyond this paper: any study that uses a period-*average* price level as its return target risks the same mechanical artifact.

multivariate combination of many individually-weak features.

The deliberate contrast is the hand-built Cochrane–Piazzesi factor (C-CP), which by construction loads entirely on the forward-rate family — the benchmark for what a genuine single-factor predictor would look like against these diffuse fits. The acceptance is broad-based and emergent, which is why the formal tests that follow find nothing to keep.

4.1.3 How hard is the forecast?

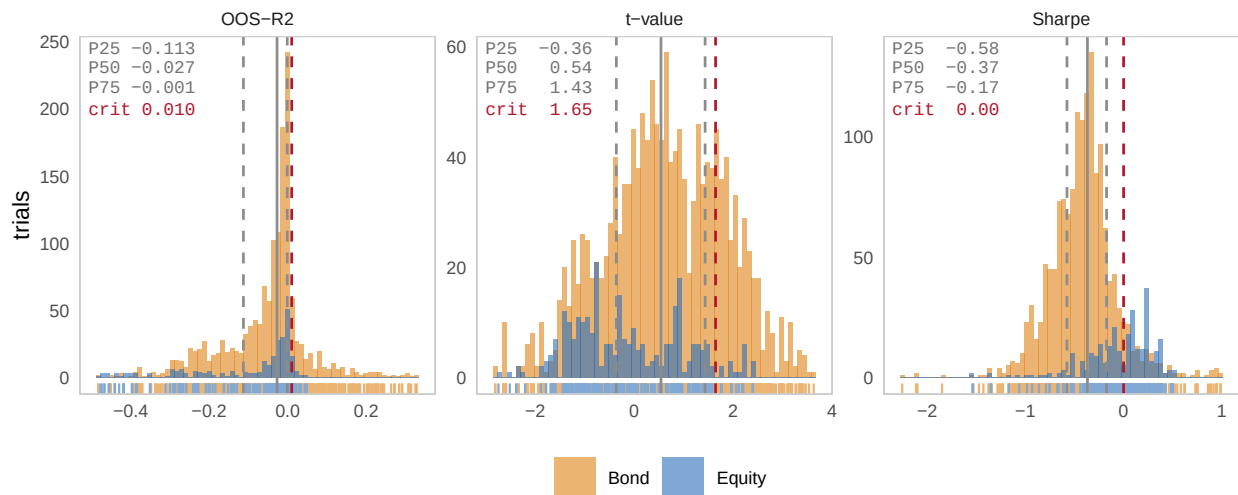


Figure 2: Distribution of the three acceptance criteria ($\star$) across the 1,806 evaluable trials (bond and equity). Dashed red marks each criterion’s bar: the 0.01 out-of-sample R^2 floor, the t -value 1.645, and (Sharpe panel) the excess over each target’s PIT-expanding median Sharpe, the bar at zero. Grey lines mark the pooled quartiles of each criterion across all evaluable trials — dashed at the lower and upper quartiles, solid at the median. A box at the left of each panel lists these pooled quartile values and the panel’s criterion. The R^2 axis is clipped at -0.5 for readability; 50 trials with more negative R^2 fall beyond it and fail the floor a fortiori. On every criterion most trials fall short, and the target-relative Sharpe is the harshest leg.

Before turning to the tests, it is worth asking the prior question the acceptance rates skip over: *how good are these forecasts in the first place?* Figure~2 sets the whole evaluable universe against the three acceptance bars at once, and the answer is a near-null on every one. The median trial’s out-of-sample R^2 is -2.7% ; only 13.1% of trials clear the one-percent floor, 19.7% clear the t -value bar, and — the harshest leg — only 15.2% beat their target’s own buy-and-hold Sharpe. Pooled across every retained equity trial the out-of-sample R^2 is -0.07 : the data-driven forecasts are, on average, *worse* than simply extrapolating the trailing historical mean.

The same verdict holds method by method. The *mean out-of-sample R^2 by estimator* panel of Annex Figure~4 averages the out-of-sample R^2 across the data-driven (Scenario-2) trials of each estimator: every shrinker’s mean is at or below zero on both targets — ridge is the least-bad and barely positive on bonds, while partial least squares, the most aggressive of the three, posts the worst average on each target. On the wide macro pool the estimators do not, on average, beat the trailing historical mean.

The two tables above list the ten strongest trials by out-of-sample R^2 for each target. Even the best are modest and concentrated: the strongest equity trial reaches only 4.9% and the strongest

Table 6: The ten strongest equity Scenario-2 trials, ranked by out-of-sample R^2 (level pools). Even the best clears only a few percent, and none survives the Part-II tests.

Method	h	Window	Sub-sample	OOS R^2 (%)	t
Ridge	3	expand	post 2008	4.9	1.20
Ridge	3	expand	full	3.4	1.12
Ridge	1	expand	post 2008	2.1	1.32
ENet	1	roll-60	pre 2008	2.1	1.46
Ridge	1	expand	full	1.8	1.28
ENet	1	roll-60	full	1.1	1.56
ENet	1	roll-60	post 2008	1.0	1.53
ENet	1	roll-48	post 2008	0.7	1.77
ENet	1	roll-48	full	0.6	1.76
ENet	1	roll-48	post 2016	0.5	2.11

Table 7: The ten strongest bond Scenario-2 trials, ranked by out-of-sample R^2 (level pools), with maturity. All ten sit in the pre-2008 sub-sample under the C-PLS estimator — a single fragile regime-and-method corner, which the multiple-testing controls of Part II do not let stand.

Maturity	Method	h	Window	Sub-sample	OOS R^2 (%)	t
2Y	PLS	3	expand	pre 2008	32.5	1.51
1Y	PLS	1	roll-48	pre 2008	30.9	2.13
1Y	PLS	3	expand	pre 2008	30.8	1.67
1Y	PLS	1	roll-60	pre 2008	29.0	2.45
2Y	PLS	1	roll-60	pre 2008	27.2	2.54
1Y	PLS	1	expand	pre 2008	24.5	2.20
2Y	PLS	1	roll-48	pre 2008	24.1	2.33
3Y	PLS	1	roll-60	pre 2008	23.6	2.18
2Y	PLS	1	roll-84	pre 2008	22.9	2.94
1Y	PLS	1	roll-84	pre 2008	22.4	2.97

bond trial 32.5% — the latter, and in fact all ten of the bond leaders, falling entirely within the pre-2008 sub-sample and the single C-PLS estimator, the tell-tale signature of a regime-specific fluke rather than a durable signal. Plotting the strongest model of each shrinkage estimator against what actually happened (Annex Figure~6) tells the same story — a timid fan of forecasts around a near-flat realized-on-forecast slope. The forecasts are hard to tell from noise; the formal tests that follow confirm it, and the figure carries their verdict alongside each fit.

The difficulty reaches all the way down to the economic families themselves — but not symmetrically across the two targets. Figure~3 grades each indicator family twice for the one-year bond and equity at $h=1$ — by its in-sample marginal R^2 (with full-sample look-ahead) and by its honest point-in-time walk-forward out-of-sample value.

On equity the collapse is total: in sample many families look predictive, yet out of sample **every one** of the 17 falls below zero. The bond short end behaves differently. There, 8 of 17 families keep a positive out-of-sample value, and 5 of them — labour, housing, monetary-policy/rates, the yield curve and growth/activity — read *higher* out of sample than in. These are precisely the real-activity and term-structure channels the bond-risk-premium literature builds on, and their survival is plausibly what lifts the one-year bond to the edge of predictability. The near-null is thus a property of the data at every altitude, from the entire universe evaluated down to the feature family with one economically coherent exception at the bond short end, which Part II puts to the test.

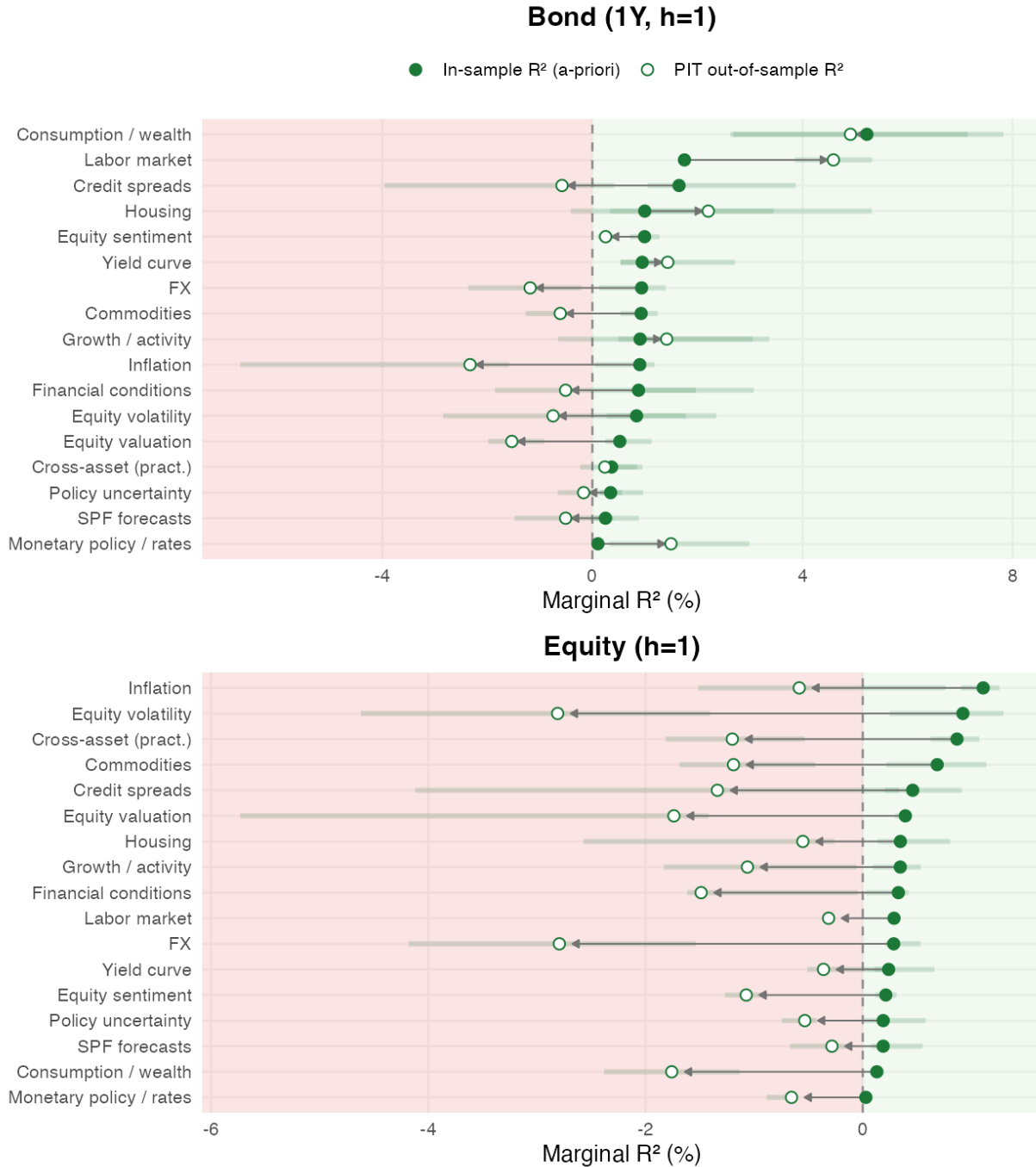


Figure 3: Where the a-priori signal lives, and what survives out of sample. Family-level marginal R^2 — in-sample (a-priori, full-sample look-ahead) versus PIT walk-forward out-of-sample — by indicator family, for the one-year bond and equity at $h=1$. Filled dots mark the in-sample marginal R^2 ; hollow dots the out-of-sample value; the arrow traces the decay from one to the other (drawn only where the gap is non-trivial, and stopping just short of the out-of-sample dot). Whiskers are the family inter-quartile range; the light-red band marks the no-go region $R^2 < 0$ and the light-green band the positive region. The gap from filled to hollow is overfit plus regime decay — on equity the decay is total, with all 17 families falling below zero out of sample, the family-level analogue of the paper’s near-null; on the one-year bond eight families keep a positive out-of-sample value and five (labour, housing, monetary policy/rates, yield curve, growth/activity) read higher out of sample than in — the real-activity and term-structure channels, and the family-level trace of the regime-confined short-end exception that Part II isolates.

4.2 Part II — The honest evaluation

Part I left essentially one acceptance to test: a short-end bond cluster (5.0% overall, in the one-to three-year maturities at $h=1$). On equity the macro data cleared the bar in 0.0% of trials, so there is nothing there to adjudicate. The bond cluster is exactly the kind of apparent signal that a sweep over 1,806 evaluable, heavily-correlated trials throws off by chance, and it counts as a result only if it survives the control of Section 2.6. Every bond trial gets the false-discovery control on its summary statistics²⁵; and, with the pointwise forecasts now retained for all five maturities, every maturity also faces the SPA — so the short end, where the cluster lives, faces both controls, and the longer maturities are SPA-tested rather than taken on the false-discovery control alone. The equity-Scenario-2 trials keep their full forecast paths, so — though none was accepted — we still put them through the SPA test and the two equity-only robustness checks, to confirm no positive remnant hides where the acceptance rule did not look.

4.2.1 The bond acceptance under false-discovery control

Per *trial* the bond signal looks real: the most significant one-year trials post t -values above 3.4 at raw p -values below 0.001, and the strongest *accepted* one-year trial (C-PLS, $h=1$, rolling-48, full sample) reaches an OOS R^2 of 0.062. Nor is it an isolated trial: 56 bond trials carry a raw $p < 0.01$ on the t -value together with a positive out-of-sample R^2 , overwhelmingly at the monthly horizon. The question is whether that *breadth* survives the size of the search. We control the false-discovery rate three ways: each maturity as its own pre-registered family, a pooled-bond family that charges for the five-maturity search, and the full evaluable universe. BH is the lenient comparison; the dependence-robust BY is the headline, because the window $\times$ sub-sample trials within a maturity are strongly correlated.

Table 8: False-discovery control by series-family — each bond maturity and equity treated identically as its own pre-registered family, then the pooled-bond and full-universe families that charge for the search. BH (lenient) leaves short-maturity survivors; the dependence-robust BY headline leaves none in any family.

Family	Trials	Accept %	min raw p	BH surv.	min BY q	BY surv.
Bond 1Y	306	15.4	< 0.001	98	0.084	0
Bond 2Y	306	4.6	0.003	0	0.960	0
Bond 3Y	306	3.6	0.007	0	1.000	0
Bond 5Y	306	1.3	0.006	0	1.000	0
Bond 10Y	306	0.0	0.006	0	1.000	0
Equity	276	0.0	0.011	0	1.000	0
Pooled bond	1530	5.0	< 0.001	0	0.524	0
Full universe (eq+bond)	1806	4.2	< 0.001	0	0.632	0

The bond acceptance is multiplicity, not discovery. Under the dependence-robust BY correction, zero trials survive in any family — the closest miss is the one-year bond in its own (most generous) family at $q = 0.084$, which the pooling penalty then buries to 0.524 (pooled bond) and 0.632 (universe); the *minimum BY-FDR q -value by family* panel of Annex Figure~4 plots the minimum BY q for every family. The lenient BH correction, which is *not* valid under this dependence, does leave 98 one-year trials standing in their own (most generous) family — but none once the pooled-bond (0) or universe (0) correction is applied — and the strong correlation across windows and sub-samples is exactly what BY corrects for, so BY is the headline and even that lone

²⁵Each trial's t -value referred to a Student- t with $\lfloor n_{\text{oos}}/h \rfloor - 1$ degrees of freedom.

short-end breadth dissolves. The diagnosis is the recurring one: real per trial, gone under honest multiple testing.²⁶

4.2.2 The equity acceptance under overlap-robust inference and SPA

Though no equity trial was accepted (0.0%, $h \in \{1, 3\}$), the equity-Scenario-2 trials retain pointwise forecasts, so we put them through the fuller battery to confirm the acceptance rule missed nothing. No $h=12$ exception is even possible: at $h=12$ the equity trial is infeasible under the effective-sample floor ($\lfloor n_{\text{oos}}/h \rfloor \geq 30$ is unattainable on this sample), so equity is tested wherever it is feasible — there is no horizon at which a positive remnant could hide.

Table 9: Equity-S2 forecast-based refinements (all 144 equity-S2 trials retain pointwise forecasts): the overlap-robust t and the in-sample efficiency check, which use the equity one-period innovation. Each row counts trials passing that filter at p or $q < 0.05$. A few survive per trial; none clears false-discovery control.

Filter	Trials passing (p or $q < 0.05$)
Overlap-robust t	4
BH-FDR (OOS)	0
BY-FDR (OOS)	0
In-sample efficiency	0
FDR (in-sample)	0

Of the 144 equity-S2 trials, 4 keep a per-trial t -value at $p < 0.05$ after the overlap penalty, and 0 reach nominal in-sample significance — yet **zero** clear BH or BY false-discovery control in either direction (every OOS q exceeds 0.05 by a wide margin). The SPA test tells the same story per (horizon $\times$ sub-sample):

No equity family clears. Of the 12 SPA families, 0 reach $p < 0.05$. The least-dead is post-2016 at $h=1$ ($T_{\text{SPA}} = 2.01$, $p = 0.106$) — nowhere near a near-miss — and every other family sits at $p \geq 0.44$. Combined with the zero FDR survivors, the equity result is an unambiguous near-null: a few trials look significant in isolation, but nothing survives honest correction.

4.2.3 Shades of grey: grading the regions

The near-null is not uniform. We grade each Scenario-2 region — target $\times$ maturity (bond) $\times$ horizon $\times$ sub-sample — by a composite of OOS- R^2 sign, t -value, and FDR/SPA survival: *nothing* (median OOS $R^2 \leq 0$ or median t -value < 1.645 , i.e. no per-trial signal), *partial* (per-trial signal present but dissolved by correction), and *half-decent* (a BY-FDR or SPA survivor). The table reports each live region’s best dependence-robust q and its SPA p wherever the forecasts were retained (equity, and all five bond maturities), so the gradient is visible row by row.

The half-decent tier has a single occupant, and it is a regime-confined exception rather than a discovery. Running the SPA across all five maturities flags only the pre-2008 one-year

²⁶Every bond maturity was forecast-retained and put through the SPA per (horizon $\times$ sub-sample), not the one-year alone. The result is the predicted single exception: two pre-2008 one-year families clear the SPA (strongest $p = 0.037$) yet fail this very BY-FDR ($q_{\text{BY}} \geq 0.137$), rest on roughly 37 common observations under the overfit-prone C-PLS estimator, and do not persist past 2008 (Part II, shades-of-grey); no longer maturity clears the SPA at any horizon or sub-sample (next-best $p = 0.097$). SPA and FDR are not nested, so a snooping-robust trial can survive a dead FDR — and exactly one regime-confined cluster does, a thin exception rather than breadth. The overlap-robust t and in-sample efficiency legs remain equity-only (they use the equity one-period innovation).

Table 10: SPA on the equity-S2 sub-family, per (horizon $\times$ sub-sample). The all-windows family pools every estimation window; the expanding-only variant appears at $h=1$, the only horizon where it differs (at $h=3$ the effective-sample floor leaves no rolling-window trial, so the two coincide). No family clears $p < 0.05$; $T_{\text{SPA}}=0$ marks a family whose best model never beats the benchmark.

Window	h	Sample	Models	Common obs	T_{SPA}	p (consistent)
all windows	1	post 2016	30	123	2.01	0.106
all windows	1	pre 2008	30	37	1.52	0.440
expanding only	1	post 2008	6	219	0.52	0.447
expanding only	1	full	6	256	0.46	0.476
all windows	3	post 2008	6	217	0.43	0.512
all windows	3	pre 2008	6	37	0.00	0.544
all windows	3	full	6	254	0.32	0.553
all windows	3	post 2016	6	121	0.00	0.684
expanding only	1	post 2016	6	123	0.00	0.738
expanding only	1	pre 2008	6	37	0.00	0.765
all windows	1	post 2008	30	219	0.52	0.799
all windows	1	full	30	256	0.46	0.838

Table 11: Shades-of-grey grading of the closest-to-surviving Scenario-2 regions, bond and equity treated alike (each bond maturity, and equity, as a series). Ranked by closeness to surviving — the BY-FDR q for bonds, the SPA p for equity. Every bond maturity is now forecast-retained, so each region carries an SPA p ; only the pre-2008 one-year bond clears it. Two pre-2008 one-year-bond regions grade half-decent: they clear the snooping-robust SPA but fail the dependence-robust BY-FDR — the single regime-confined exception discussed in the text. Every other region grades partial or nothing.

Series	Region	Accept %	med R2	med t	best BY- q	SPA p	Grade
Bond 1Y	h1 / full	40.0	0.038	3.09	0.084	0.171	partial
Bond 1Y	h1 / post 2008	16.7	-0.027	2.37	0.084	0.583	nothing
Bond 1Y	h1 / post 2016	33.3	0.002	2.09	0.084	0.379	partial
Equity	h1 / post 2016	0.0	-0.057	-0.92	1.000	0.106	nothing
Bond 1Y	h1 / pre 2008	60.0	0.183	2.25	0.137	0.042	half-decent
Equity	h1 / pre 2008	0.0	-0.051	-0.75	1.000	0.440	nothing
Equity	h1 / post 2008	0.0	-0.032	0.84	1.000	0.447	nothing
Equity	h1 / full	0.0	-0.033	0.80	1.000	0.475	nothing
Bond 1Y	h3 / pre 2008	33.3	0.203	2.00	0.503	0.037	half-decent
Equity	h3 / post 2008	0.0	-0.034	1.19	1.000	0.512	nothing
Equity	h3 / pre 2008	0.0	-0.188	-2.38	1.000	0.544	nothing
Equity	h3 / full	0.0	-0.032	1.11	1.000	0.553	nothing

bond: 2 regions clear the snooping-robust test ($h=1$ and $h=3$, the stronger at $p = 0.037$), and no longer maturity clears it at any horizon or sub-sample (next-best $p = 0.097$). But that very cluster *fails* the dependence-robust BY-FDR ($q = 0.137$ at $h=1$, 0.503 at $h=3$), rests on a short common evaluation axis (37 months — about twelve effective observations at $h=3$, below the $n_{\text{eff}} \geq 30$ floor the rest of the paper imposes), is driven by the overfit-prone C-PLS estimator (every one of the ten strongest bond trials is pre-2008 C-PLS), and evaporates after 2008 (SPA $p > 0.5$). It is exactly the *single thin exception* — snooping-robust at one point yet dead under the breadth control, and confined to one early regime — that the non-nesting of SPA and FDR anticipated, not the durable short-end signal the per-trial t -values seemed to promise. On equity the half-decent tier is genuinely empty: the closest family is post-2016 at $h=1$ (SPA $p = 0.106$). Every region not shown — including all of $h=12$, where the floor leaves nothing feasible — grades *partial* or *nothing*.

5 Conclusions

In this research, we have approached the bond and equity excess return predictability from a practitioner perspective. This pathway incorporates technical rigor — of a rather boring nature — centered around data collection, pre-processing and the development of what should be considered a responsible forecasting pipeline. Throughout, the strict PIT construction means these are honest OOS numbers: every forecast at date T uses only data released by T , and no result is inflated by vintage or release-timing leakage. The near-null is not a disappointment but the point — it is what an honest evaluation of monthly macro-return predictability looks like, and it is consistent with Welch and Goyal (2008). The deliverable for downstream factor-model work is the panel and its accessors, together with two durable lessons: construct the target before trusting the signal, and control for multiple testing before believing a grid of acceptances.

What this leaves is a clear and largely negative answer. On strictly public, point-in-time data, monthly predictability of the aggregate equity and bond risk premium is, for practical purposes, dead on arrival: the in-sample signal that motivates the search does not survive an honest out-of-sample evaluation, and the handful of acceptances a wide grid produces is the residue of multiple testing, not discovery. The two targets fail slightly differently. Equity is a clean null — every indicator family, without exception, loses its apparent edge out of sample. Bonds keep a faint but economically coherent trace: the real-activity and term-structure families the bond-risk-premium literature has always pointed to are the few that retain positive out-of-sample value at the short maturity, and are plausibly what lifts the one-year bond to the edge of predictability. Yet that trace is confined to one maturity, one horizon, and the pre-2008 regime, and it does not survive dependence-robust false-discovery control. A single regime-bound exception is not a reversal of the null.

Annex

A.1 Supplementary figures

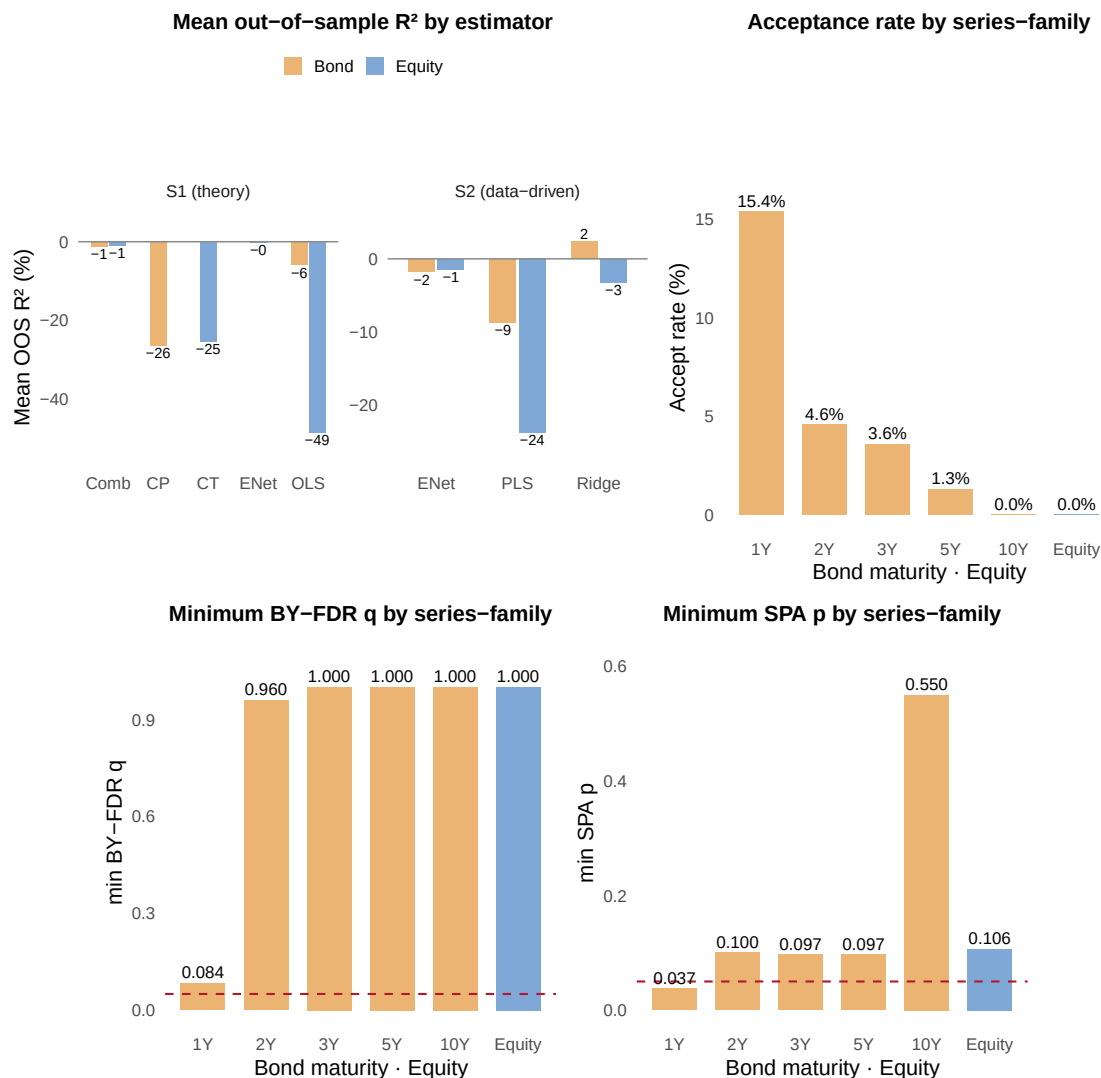


Figure 4: Supplementary diagnostics, four panels; the lower three share one *series-family* axis — the five bond maturities and equity, each treated as a single series — so both targets are shown identically throughout. *Mean out-of-sample R^2 by estimator* averages the OOS R^2 (%) across each estimator’s trials, split by target and by scenario (S1 theory pools, S2 data-driven); every estimator’s average sits at or below zero on both targets in both scenarios — the unconstrained OLS-class theory configurations overfit most violently out of sample, while the shrinkage class barely dips below zero (ridge least-bad, just positive on bonds). *Acceptance rate by series-family* gives the acceptance rate (rule $\star$, evaluable basis); it declines monotonically in bond duration, concentrated at the one-year maturity and zero at the ten-year end and on equity. *Minimum BY-FDR q by series-family* plots the smallest dependence-robust q -value within each series-family; none clears the $q < 0.05$ line (dashed), the closest the one-year bond. *Minimum SPA p by series-family* plots the smallest SPA consistent p (across horizon and sub-sample) within each series-family; only the one-year bond dips below the $p < 0.05$ line (dashed), every longer maturity and equity sitting well above — the short end is the lone exception across the whole curve.

Scenario 1 – academic theory

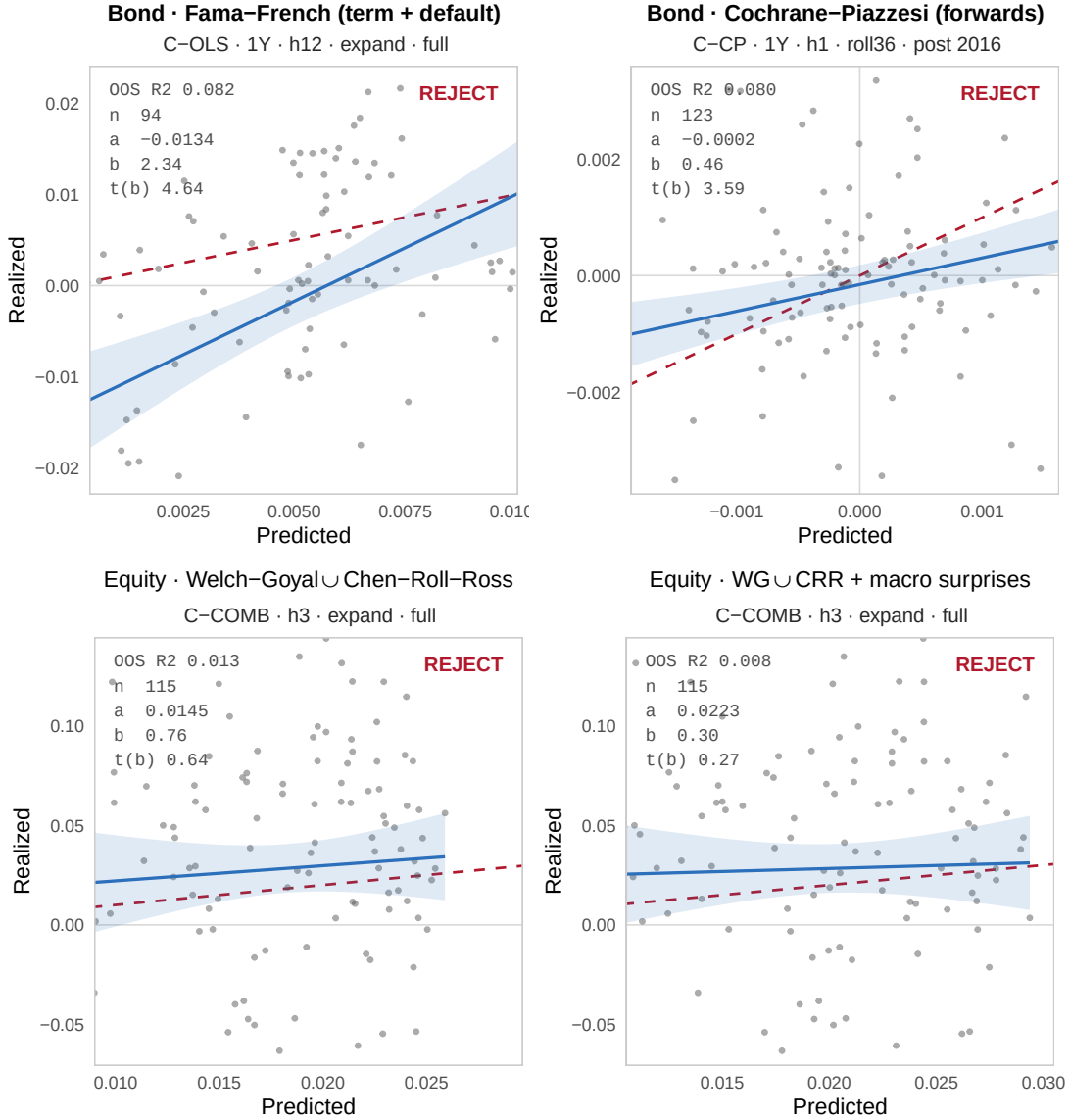


Figure 5: Scenario 1 (academic theory). Predicted versus realized excess returns for the single strongest trial (highest pooled OOS R^2 , ≥ 24 points) of each classic-literature pool: the Fama–French (1989) term-and-default and Cochrane–Piazzesi forward-rate bond pools on the top row, the Welch–Goyal $\cup$ Chen–Roll–Ross equity pool (and its surprise-augmented variant) below. The Scenario-1 counterpart to the data-driven bridge (Figure 6): the blue line is the realized-on-forecast fit (95% CI), the red dashed line the perfect forecast, the box reports OOS R^2 , n and the fit’s intercept a , slope b and slope t -value $t(b)$, and the subtitle gives the trial’s identity. The bold tag is the acceptance verdict (rule $\star$) — every pool is a REJECT, and across the full sweep the four pools accept exactly zero trials. Even cherry-picked the equity canons barely move the forecast ($R^2 \approx 1\%$) and the bond pools reach $\approx 8\%$ only in an isolated window-and-regime trial — the same regime-specific fragility Part II isolates, now visible inside the theory replications.

Scenario 2 – data-driven

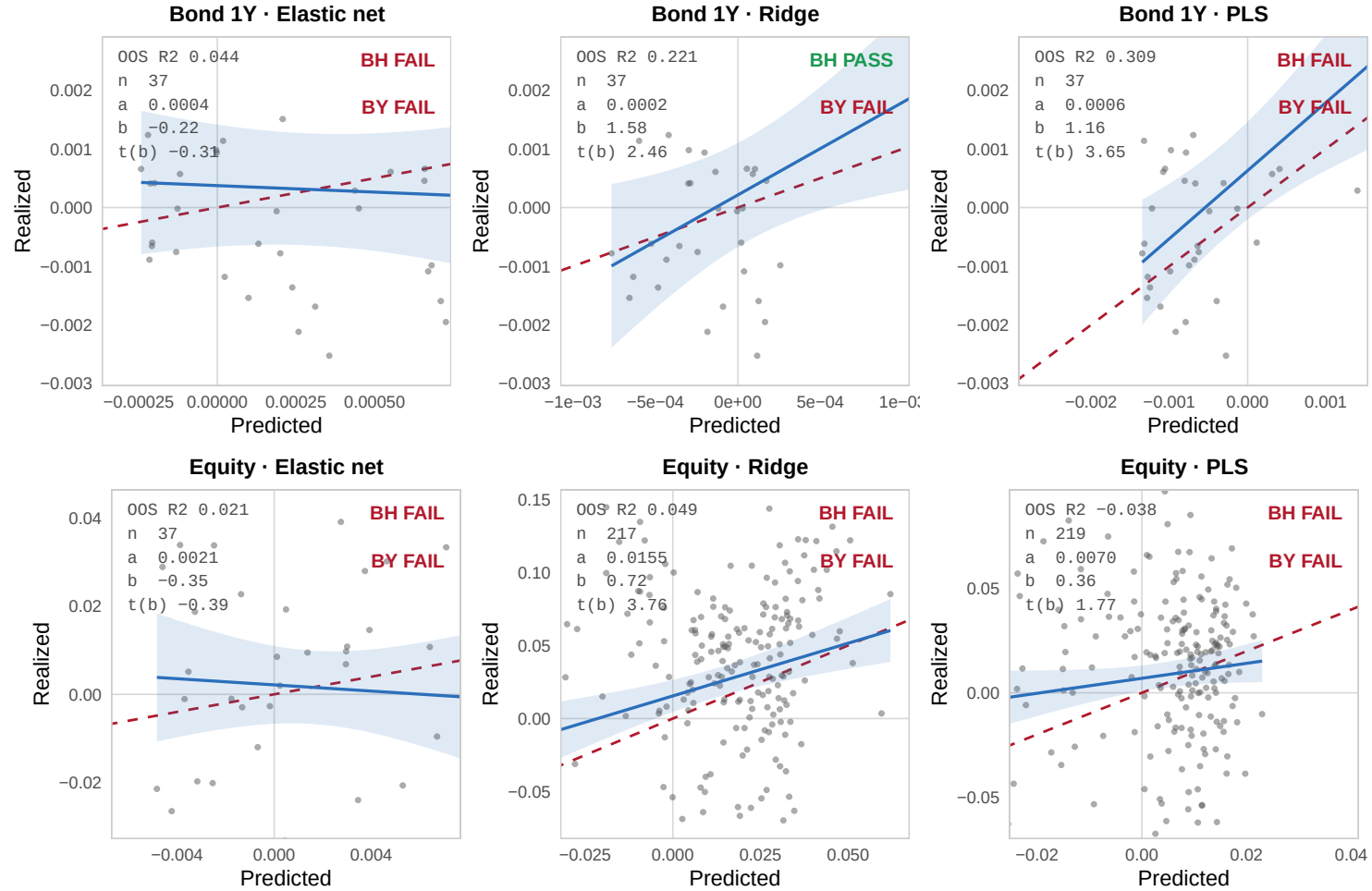


Figure 6: Scenario 2 (data-driven). Predicted versus realized excess returns for the strongest model of each shrinkage estimator and target — the one-year bond across the top row, equity across the bottom, columns elastic net / ridge / PLS. Each panel is zoomed about its own median; the blue line is the realized-on-forecast fit (95% CI), the red dashed line the perfect forecast (slope one), and the box reports OOS R^2 and n together with that fit's intercept a , slope b and slope t -value $t(b)$. The bold tag is the model's own-family (most generous) false-discovery verdict, BH and BY at $q < 0.05$. Even the strongest fits barely track the outcome: only the one-year-bond ridge model clears lenient BH, and none clears BY.

A.2 Data revisions across the vintage corpus

This appendix audits *how much* the panel’s revised series are revised, across every one. The universe is the 132 VINTAGE-tier, non-forecast series (FRED/ALFRED and the Philadelphia Fed Real-Time Data Set); SPF survey rounds are excluded as forecasts, not revisions. For each series we compare the first release with the latest value on the year-over-year log-growth the estimators consume (§2.3) — which cancels base-year rebasing — and summarise it by the noise-to-signal ratio $NS = \text{sd}(\text{revision}) / \text{sd}(\text{latest growth})$ (Croushore and Stark 2001; Aruoba 2008). Two series that take non-positive values (real net exports and a net-saving series) are dropped and named rather than silently. Revisions are pervasive (median $NS = 0.29$; 25% above 0.5) and two-sided (53% upward) — they cannot be undone by assuming a fixed first-release bias.

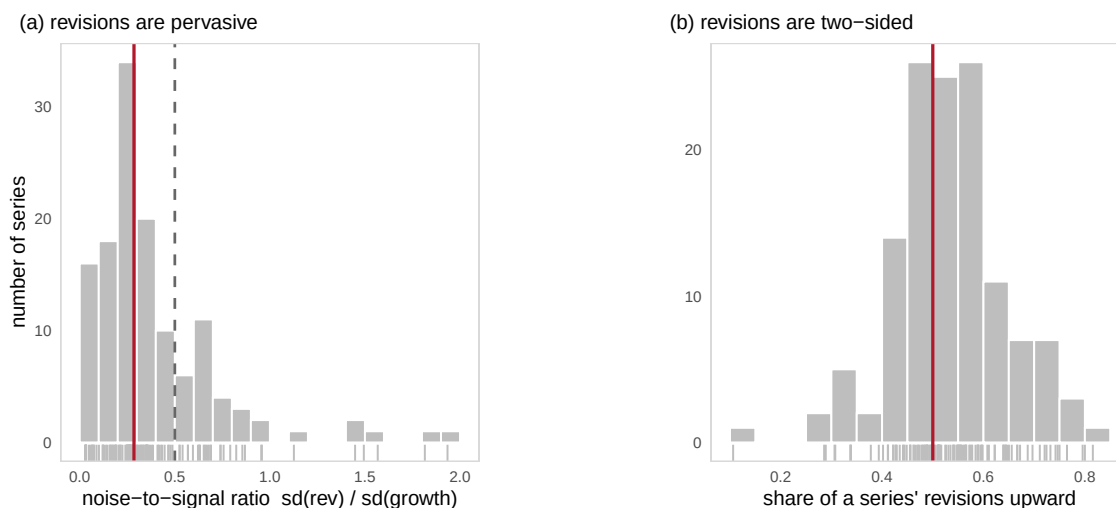


Figure 7: Data revisions across the vintage corpus. Each series’ first-release versus latest year-over-year growth, across every genuinely-revised series in the panel. (a) Distribution of the noise-to-signal ratio (standard deviation of the revision over standard deviation of the latest growth): revisions are pervasive, the median marked in red and the 0.5 reference dashed, with a right tail past 1 where a revision is as large as the signal itself. (b) Distribution of the share of a series’ revisions that are upward, centred on 0.5 (red): revisions are two-sided and cannot be undone by assuming first prints are uniformly too high or too low. Exact counts are in the Section 3 footnote and the per-series table below.

Table 12: Revision magnitude by economic family: noise-to-signal ratio, median absolute year-over-year-growth revision (percentage points), and share of revisions that are upward.

Family	Series	Median N/S	Median abs rev (pp)	Pct upward
INCOMES	18	0.45	3.26	57
GROWTH	19	0.35	0.93	55
CONSUMPTION_WEALTH	43	0.33	0.80	53
LABOR_MARKET	15	0.26	0.36	46
MONEY	6	0.16	0.35	51
INFLATION	22	0.14	0.13	56
HOUSING	7	0.12	1.80	52

Table 13: Every genuinely-revised series, sorted by noise-to-signal ratio. n = usable reference months; Vint = vintages on file; Med abs rev = median absolute year-over-year-growth revision (percentage points); N/S = noise-to-signal ratio; Pct up = share of revisions that are upward.

Series	Family	Frq	n	Vint	Med abs rev	N/S	Pct up
RTDSM_RENTI	CONSUMPTION_WEALTH	Q	235	242	5.29	1.93	51
RTDSM_YNIPAI	INCOMES	Q	45	44	5.95	1.81	71
RTDSM_TRANPF	CONSUMPTION_WEALTH	Q	244	242	4.26	1.57	66
RTDSM_RATESAV	CONSUMPTION_WEALTH	Q	230	242	10.13	1.49	55
RTDSM_NPSAV	CONSUMPTION_WEALTH	Q	230	242	10.55	1.45	57
RTDSM_YNSD	CONSUMPTION_WEALTH	Q	21	85	66.81	1.13	29
RTDSM_YNCTAX	INCOMES	Q	45	44	7.24	0.96	47
RTDSM_NCPROFATW	INCOMES	Q	182	181	6.13	0.95	57
RTDSM_RINVCHI	GROWTH	Q	149	242	41.93	0.87	50
RTDSM_YNRINC	CONSUMPTION_WEALTH	Q	45	44	2.53	0.86	56
RTDSM_ULC	GROWTH	Q	111	110	0.93	0.82	48
RTDSM_PROPI	CONSUMPTION_WEALTH	Q	244	242	2.66	0.79	55
RTDSM_DIV	CONSUMPTION_WEALTH	Q	244	242	4.19	0.76	60
RTDSM_YNSWS	LABOR_MARKET	Q	45	44	1.18	0.74	51
RTDSM_YNTRPAY	INCOMES	Q	45	44	9.12	0.74	49
RTDSM_NCONSNP	CONSUMPTION_WEALTH	Q	68	67	3.15	0.69	46
RTDSM_RGSL	GROWTH	Q	243	242	1.08	0.69	53
RTDSM_RCONSNP	CONSUMPTION_WEALTH	Q	68	67	2.70	0.68	49
RTDSM_RGF	GROWTH	Q	243	242	1.28	0.67	56
RTDSM_YNCPRFATW	INCOMES	Q	45	44	4.38	0.66	58
RTDSM_RG	GROWTH	Q	243	242	0.89	0.66	51
RTDSM_M1	MONEY	M	734	242	0.25	0.65	65
RTDSM_RCONSNPM	CONSUMPTION_WEALTH	M	209	201	2.99	0.63	52
RTDSM_NCPROFAT	INCOMES	Q	243	242	4.64	0.63	54
RTDSM_NCONSNPM	CONSUMPTION_WEALTH	M	209	201	3.13	0.63	49
RTDSM_POP	LABOR_MARKET	M	339	330	0.00	0.62	11
RTDSM_OLI	CONSUMPTION_WEALTH	Q	244	242	1.49	0.60	51
RTDSM_OPH	GROWTH	Q	111	110	0.58	0.57	54
RTDSM_PINTI	CONSUMPTION_WEALTH	Q	244	242	2.82	0.57	56
RTDSM_YNPINCW	CONSUMPTION_WEALTH	Q	45	44	1.59	0.54	51
RTDSM_RCONND	CONSUMPTION_WEALTH	Q	243	242	0.74	0.53	50
RTDSM_PINTPAID	CONSUMPTION_WEALTH	Q	244	242	3.41	0.52	52
RTDSM_YNOSP	INCOMES	Q	45	44	1.52	0.49	62
RTDSM_YNOS	INCOMES	Q	45	44	1.56	0.48	62
RTDSM_YNUCPRFW	INCOMES	Q	45	44	20.60	0.47	29
RTDSM_YNCOMPEP	LABOR_MARKET	Q	45	44	0.85	0.47	51
RTDSM_YNWS	LABOR_MARKET	Q	45	44	0.85	0.45	51
RTDSM_YNCPRFW	INCOMES	Q	45	44	3.03	0.43	56
RTDSM_RCONS	CONSUMPTION_WEALTH	Q	243	242	0.53	0.43	60
RTDSM_RCONNDM	CONSUMPTION_WEALTH	M	338	330	0.69	0.42	39
RTDSM_RINVB	GROWTH	Q	243	242	1.55	0.41	57
RTDSM_PCONSNPM	INFLATION	M	209	201	1.11	0.41	31
RTDSM_YNDPAID	INCOMES	Q	44	44	12.83	0.39	75
RTDSM_ROUTPUT	GROWTH	Q	243	242	0.52	0.38	65
RTDSM_NDPI	CONSUMPTION_WEALTH	Q	244	242	0.83	0.37	64
RTDSM_WSD	LABOR_MARKET	Q	244	242	0.68	0.36	61
RTDSM_HS	LABOR_MARKET	M	666	656	0.40	0.36	45
RTDSM_YNTAXR	INCOMES	Q	45	44	0.99	0.36	80
RTDSM_YRGDI	INCOMES	Q	86	85	0.62	0.35	64
RTDSM_IPM	GROWTH	M	741	736	1.35	0.35	46
RTDSM_PCONSNP	INFLATION	Q	68	67	1.15	0.35	34
RTDSM_IPT	GROWTH	M	746	736	1.09	0.35	47
GDPC1	GROWTH	Q	141	414	0.44	0.35	59
RTDSM_REX	GROWTH	Q	243	242	1.26	0.35	69
RTDSM_NPI	CONSUMPTION_WEALTH	Q	244	242	0.76	0.34	67
RTDSM_YNCFCP	INCOMES	Q	45	44	0.49	0.34	44
RTDSM_RCON	CONSUMPTION_WEALTH	Q	243	242	0.41	0.33	60
RTDSM_CUM	GROWTH	M	571	561	0.84	0.33	49
RTDSM_PTAX	CONSUMPTION_WEALTH	Q	244	242	1.36	0.32	59
A229RX0	CONSUMPTION_WEALTH	M	202	191	1.07	0.32	64
HSN1F	HOUSING	M	330	317	4.27	0.31	43
RTDSM_RIMP	GROWTH	Q	243	242	1.16	0.30	56
RTDSM_CUT	GROWTH	M	524	514	0.82	0.30	46
RTDSM_PCONSHHM	INFLATION	M	209	201	0.20	0.30	75
RTDSM_PCONSHH	INFLATION	Q	68	67	0.19	0.29	76
RTDSM_RCONDM	CONSUMPTION_WEALTH	M	338	330	1.21	0.28	43

Table 13: Every genuinely-revised series, sorted by noise-to-signal ratio. n = usable reference months; Vint = vintages on file; Med abs rev = median absolute year-over-year-growth revision (percentage points); N/S = noise-to-signal ratio; Pct up = share of revisions that are upward. (continued)

Series	Family	Frq	n	Vint	Med abs rev	N/S	Pct up
RTDSM_YNCFC	INCOMES	Q	45	44	0.42	0.28	51
RTDSM_NBRECBASE	MONEY	M	336	112	0.44	0.28	51
GDP	GROWTH	Q	141	410	0.45	0.28	67
RTDSM_RCONGM	CONSUMPTION_WEALTH	M	209	201	0.76	0.28	41
RTDSM_YNGDI	INCOMES	Q	86	85	0.75	0.28	64
RTDSM_SSCONTRIB	LABOR_MARKET	Q	244	242	0.91	0.27	42
RTDSM_PCONX	INFLATION	Q	122	121	0.15	0.27	70
RTDSM_RINVRESID	HOUSING	Q	243	242	1.84	0.27	58
PCEPILFE	INFLATION	M	318	311	0.17	0.27	80
RTDSM_RCONHH	CONSUMPTION_WEALTH	Q	68	67	0.44	0.26	51
RTDSM_RCONG	CONSUMPTION_WEALTH	Q	68	67	0.76	0.26	43
RTDSM_LFC	LABOR_MARKET	M	339	330	0.04	0.26	55
RTDSM_RCONSHH	CONSUMPTION_WEALTH	Q	68	67	0.44	0.26	56
TCU	GROWTH	M	364	380	0.72	0.26	44
RTDSM_RCONSM	CONSUMPTION_WEALTH	M	338	330	0.38	0.26	49
RTDSM_NOUTPUT	GROWTH	Q	243	242	0.48	0.26	72
RTDSM_RCOND	CONSUMPTION_WEALTH	Q	243	242	1.15	0.26	53
RTDSM_RCONHHM	CONSUMPTION_WEALTH	M	209	201	0.45	0.26	50
RTDSM_RCONM	CONSUMPTION_WEALTH	M	338	330	0.38	0.25	46
RTDSM_RCONSHHM	CONSUMPTION_WEALTH	M	209	201	0.46	0.25	53
PCEC96	CONSUMPTION_WEALTH	M	219	559	0.40	0.25	50
RTDSM_NCONSHH	CONSUMPTION_WEALTH	Q	68	67	0.50	0.25	59
RTDSM_NCON	CONSUMPTION_WEALTH	Q	244	242	0.46	0.25	64
INDPRO	GROWTH	M	1202	1218	1.31	0.25	57
RTDSM_NCONSHHM	CONSUMPTION_WEALTH	M	209	201	0.51	0.24	61
RTDSM_M2	MONEY	M	668	220	0.27	0.22	50
RTDSM_NCONHH	CONSUMPTION_WEALTH	Q	68	67	0.47	0.22	57
RTDSM_NCONHHM	CONSUMPTION_WEALTH	M	209	201	0.51	0.21	54
RTDSM_NCONGM	CONSUMPTION_WEALTH	M	209	201	0.70	0.21	41
RTDSM_TRANR	CONSUMPTION_WEALTH	Q	244	242	0.80	0.20	54
RTDSM_PIMP	INFLATION	Q	243	242	0.57	0.19	45
RTDSM_NCONG	CONSUMPTION_WEALTH	Q	68	67	0.74	0.19	41
RTDSM_H	LABOR_MARKET	M	666	656	0.36	0.19	47
RTDSM_YNCFCG	INCOMES	Q	45	44	0.36	0.19	67
RTDSM_P	INFLATION	Q	243	242	0.25	0.18	55
PAYEMS	LABOR_MARKET	M	862	854	0.25	0.18	59
PCEPI	INFLATION	M	318	311	0.13	0.17	74
RTDSM_YNGSUB	INCOMES	Q	45	44	3.48	0.17	38
RTDSM_HG	LABOR_MARKET	M	666	656	0.55	0.16	50
RTDSM_EMPLOY	LABOR_MARKET	M	746	736	0.24	0.16	58
RTDSM_PCON	INFLATION	Q	244	242	0.17	0.16	59
RTDSM_PCONHHM	INFLATION	M	209	201	0.14	0.15	73
RTDSM_PCONHH	INFLATION	Q	68	67	0.13	0.14	72
HOUST	HOUSING	M	795	785	1.80	0.13	55
RTDSM_YPDGDP	INFLATION	Q	45	44	0.10	0.12	64
PERMIT	HOUSING	M	331	612	2.26	0.12	82
RTDSM_HSTARTS	HOUSING	M	699	699	1.73	0.12	55
RTDSM_LFPART	LABOR_MARKET	M	339	330	0.04	0.10	49
RTDSM_NBRBASE	MONEY	M	415	133	0.56	0.09	48
USSTHPI	HOUSING	Q	65	63	0.31	0.08	31
RTDSM_PPPIX	INFLATION	M	340	330	0.06	0.07	55
UNRATE	LABOR_MARKET	M	803	794	0.36	0.07	34
RTDSM_RUC	LABOR_MARKET	M	733	242	0.28	0.07	34
RTDSM_PCONGM	INFLATION	M	209	201	0.11	0.06	40
RTDSM_PCONG	INFLATION	Q	68	67	0.10	0.06	44
RTDSM_PPPI	INFLATION	M	340	330	0.11	0.05	47
CPILFESL	INFLATION	M	362	371	0.02	0.03	49
CSUSHPINS	HOUSING	M	147	138	0.10	0.03	42
RTDSM_PCPPIX	INFLATION	M	339	330	0.02	0.03	48
RTDSM_CPI	INFLATION	M	388	127	0.03	0.03	48
RTDSM_BASEBASE	MONEY	M	402	124	0.08	0.03	50
RTDSM_PCPI	INFLATION	M	339	330	0.02	0.03	48
RTDSM_TRBASE	MONEY	M	437	133	0.57	0.03	43
CPIAUCSL	INFLATION	M	651	664	0.04	0.03	50

What a revised-data shortcut would buy. A complement to *how much* series are revised is *what using the revised values buys a backtest that should not have had them*. We re-run the shrinkage class (ridge, PLS, elastic net) on the one-year bond and the equity index under a *vintage-leaked* panel — each reference month carries its latest-revision value but keeps its original release date, so only revision content changes and the comparison is otherwise identical (the point-in-time leg reproduces the headline trials exactly, $\max |\Delta| = 0$). Table 14 reports the bond trials, where the panel’s only surviving signal lives. The leak inflates essentially every statistic — out-of-sample R^2 rises in 7 of the 9 trials and the Clark–West t in 7 — most sharply for the pre-2008 short-end cluster: the one-year bond’s best tradable Sharpe climbs from 0.82 to 1.19 (partial least squares, $h=1$) and one elastic-net R^2 inflates roughly eight-fold ($0.007 \rightarrow 0.055$). Yet not one verdict changes — the same three pre-2008 short-end trials accept under both panels, every other trial rejects under both, and the equity trials (all rejected) behave identically. Vintage leakage does not manufacture discoveries the way the release-timing artifact of Part I does; it exaggerates the effect size of the discovery already there — which is why the vintage record matters for honest magnitudes even where it would not have overturned this paper’s near-null verdict.

Table 14: Vintage-leak counterfactual, one-year bond. Each trial’s point-in-time value versus a leaked panel that uses final revisions at fixed release timing; OOS- R^2 , Clark–West t and signal Sharpe shown as PIT→leaked, and $\star$ marks acceptance (rule $\star$) as PIT/leaked. The accept set is invariant ($\star$ column identical both sides) while magnitudes inflate.

Sample	h	Est.	OOS- R^2	CW t	Sharpe	$\star$
full	1	Ridge	$0.054 \rightarrow 0.066$	$2.28 \rightarrow 2.44$	$0.30 \rightarrow 0.28$	−/−
full	1	PLS	$0.003 \rightarrow 0.024$	$2.01 \rightarrow 2.22$	$0.04 \rightarrow 0.11$	−/−
full	1	ENet	$-0.003 \rightarrow -0.006$	$-0.02 \rightarrow 1.11$	$0.13 \rightarrow 0.13$	−/−
pre-08	1	Ridge	$0.168 \rightarrow 0.179$	$2.42 \rightarrow 2.55$	$0.75 \rightarrow 0.85$	$\star/\star$
pre-08	1	PLS	$0.245 \rightarrow 0.265$	$2.20 \rightarrow 2.38$	$0.82 \rightarrow 1.19$	$\star/\star$
pre-08	1	ENet	$0.007 \rightarrow 0.055$	$2.33 \rightarrow 2.04$	$0.38 \rightarrow 0.38$	−/−
pre-08	3	Ridge	$0.205 \rightarrow 0.222$	$2.01 \rightarrow 2.14$	$0.44 \rightarrow 0.44$	−/−
pre-08	3	PLS	$0.308 \rightarrow 0.360$	$1.67 \rightarrow 1.82$	$1.01 \rightarrow 1.01$	$\star/\star$
pre-08	3	ENet	$0.006 \rightarrow 0.000$	$2.00 \rightarrow 1.89$	$0.44 \rightarrow 0.44$	−/−

A.3 Features effectively used in a model

Every feature below enters at least one prediction pool: the theory-shrunk Scenario-1 pools (**S1-eq** = Welch–Goyal $\cup$ Chen–Roll–Ross; **S1-bond/FF** = Fama–French term and default spreads; **S1-bond/CP** = the Cochrane–Piazzesi forward-rate inputs), the 114-feature all-variables data-driven Scenario-2 pool (**S2**, of which 13 are the practitioner **dashboard** constructs of the M1 sensitivity), or the surprise constructs. The table is audited for exact coverage — every listed feature is used, and every used feature is listed (0 stale / 0 missing, verified in `working_paper/pipeline/build_annex_table.R`).

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership).

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
1	DCOILBRENTU	Crude Oil Prices: Brent - Europe (daily, USD/bbl)	COMMODITIES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Bernanke (2016); EIA	S2
2	GOLD	COMEX Gold front-month (Yahoo GC=F, month-end total-return proxy)	COMMODITIES	Yahoo Finance GC=F (frozen: sources/equity/gcf_monthly_yahoo_2026-06-04.csv)	level/spread, trailing 60-mo z-score	Standard practitioner	S2
3	A229RX0	Real Disposable Personal Income (monthly, SAAR)	CONSUMPTION WEALTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Lettau & Ludvigson (2001)	S2
4	PCEC96	Real Personal Consumption Expenditures (monthly, SAAR)	CONSUMPTION WEALTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Lettau & Ludvigson (2001)	S2
5	AAA	Moody’s Seasoned Aaa Corporate Bond Yield (monthly average)	CREDIT SPREADS	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Fama & French (1989)	S2
6	AGG	iShares Core U.S. Aggregate Bond ETF (US-Agg total-return proxy)	CREDIT SPREADS	Yahoo Finance AGG (frozen: sources/equity/agg_monthly_yahoo_2026-06-04.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (iShares)	S2
7	BAA	Moody’s Seasoned Baa Corporate Bond Yield (monthly average)	CREDIT SPREADS	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Fama & French (1989)	S2
8	DEFAULT_SPREAD	Default spread (Baa minus Aaa)	CREDIT SPREADS	FRED (BAA, AAA)	derived spread (level); 60-mo z-score	Fama & French (1989)	S1-eq, S1-bond/FF

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
9	HYG	iShares iBoxx High Yield Corp Bond ETF (HY total-return proxy)	CREDIT SPREADS	Yahoo Finance HYG (frozen: sources/equity/hyg_monthly_yahoo_2026-06-06.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (iShares)	S2
10	LQD	iShares iBoxx Investment Grade Corp Bond ETF (IG total-return proxy)	CREDIT SPREADS	Yahoo Finance LQD (frozen: sources/equity/lqd_monthly_yahoo_2026-06-06.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (iShares)	S2
11	DASH_CURVE	Practitioner dashboard construct (DASH_CURVE)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
12	DASH_DEFAULT	Practitioner dashboard construct (DASH_DEFAULT)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
13	DASH_PMI_PROXY	Practitioner dashboard construct (DASH_PMI_PROXY)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
14	DASH_REAL2Y	Practitioner dashboard construct (DASH_REAL2Y)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
15	X_EMDM_MOM	Practitioner dashboard construct (X_EMDM_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
16	X_EURJPY_MOM	Practitioner dashboard construct (X_EURJPY_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
17	X_EURJPY_VAL	Practitioner dashboard construct (X_EURJPY_VAL)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
18	X_HYIG_MOM	Practitioner dashboard construct (X_HYIG_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
19	X_INTLUS_MOM	Practitioner dashboard construct (X_INTLUS_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). *(continued)*

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
20	X_OILGOLD_MOM	Practitioner dashboard construct (X_OILGOLD_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
21	X_OILGOLD_VAL	Practitioner dashboard construct (X_OILGOLD_VAL)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
22	X_STKBOND_MOM	Practitioner dashboard construct (X_STKBOND_MOM)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
23	X_STKBOND_VAL	Practitioner dashboard construct (X_STKBOND_VAL)	DASHBOARD DERIVED	constructed (TRP dashboard)	level construct, 60-mo z-score	T. Rowe Price (practitioner dashboard)	S2, dashboard
24	AAII_BULLBEAR	AAII Investor Sentiment Survey — Bull-Bear Spread	EQUITY SENTIMENT	aaai.com	level/spread, trailing 60-mo z-score	Hull & Qiao (2017)	S2
25	WURGLER_SENT	Baker-Wurgler Investor Sentiment Index (PCA, orthogonalized)	EQUITY SENTIMENT	pages.stern.nyu.edu	level/spread, trailing 60-mo z-score	Baker & Wurgler (2006); Hull & Qiao (2017)	S2
26	EEM	iShares MSCI Emerging Markets ETF (EM-equity total-return proxy)	EQUITY VALUATION	Yahoo Finance EEM (frozen: sources/equity/eem_monthly_yahoo_2026-06-04.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (iShares)	S2
27	EFA	iShares MSCI EAFE ETF (DM-ex-US equity total-return proxy)	EQUITY VALUATION	Yahoo Finance EFA (frozen: sources/equity/efa_monthly_yahoo_2026-06-04.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (iShares)	S2
28	SHILLER_CAPE	Shiller Cyclically Adjusted P/E (CAPE / PE10)	EQUITY VALUATION	shillerdata.com	level/spread, trailing 60-mo z-score	Campbell & Shiller (1988)	S2
29	SHILLER_PRICE	Shiller S&P Composite Stock Price Index (nominal monthly level)	EQUITY VALUATION	shillerdata.com	level/spread, trailing 60-mo z-score	Campbell & Shiller (1988)	S2
30	SP500TR	S&P 500 Total Return Index (month-end close, dividends reinvested)	EQUITY VALUATION	Yahoo Finance ^SP500TR (frozen: sources/equity/sp500tr_monthly_yahoo_2026-05-31.csv)	level/spread, trailing 60-mo z-score	Standard practitioner (S&P)	S2

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
31	GVZCLS	CBOE GVZ — Gold ETF implied volatility (GLD)	EQUITY VOLATILITY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2
32	OVXCLS	CBOE OVX — Crude Oil ETF implied volatility (USO)	EQUITY VOLATILITY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2
33	SKEW	CBOE SKEW — S&P 500 tail-risk index	EQUITY VOLATILITY	cdn.cboe.com	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2
34	VIXCLS	CBOE VIX (S&P 500 30-day implied volatility, daily close)	EQUITY VOLATILITY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2, S1-eq
35	VVIXCLS	CBOE VVIX — Vol-of-VIX (implied volatility of VIX itself)	EQUITY VOLATILITY	cdn.cboe.com	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2
36	VXEEMCLS	CBOE VXEEM — Emerging- Markets ETF implied volatility (EEM)	EQUITY VOLATILITY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	CBOE (practitioner)	S2
37	ANFCI	Chicago Fed Adjusted National Financial Conditions Index	FINANCIAL CONDITIONS COMPOSITE	chicagofed.org	level/spread, trailing 60-mo z-score	Brave & Butters (2011)	S2
38	KCFSI	Kansas City Fed Financial Stress Index	FINANCIAL CONDITIONS COMPOSITE	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Hakkio & Keeton (2009)	S2
39	NFCI	Chicago Fed National Financial Conditions Index	FINANCIAL CONDITIONS COMPOSITE	chicagofed.org	level/spread, trailing 60-mo z-score	Brave & Butters (2011)	S2
40	NFCICREDIT	Chicago Fed NFCI Credit Subindex	FINANCIAL CONDITIONS COMPOSITE	chicagofed.org	level/spread, trailing 60-mo z-score	Brave & Butters (2011)	S2
41	NFCILEVERAGE	Chicago Fed NFCI Leverage Subindex	FINANCIAL CONDITIONS COMPOSITE	chicagofed.org	level/spread, trailing 60-mo z-score	Brave & Butters (2011)	S2
42	NFCIRISK	Chicago Fed NFCI Risk Subindex	FINANCIAL CONDITIONS COMPOSITE	chicagofed.org	level/spread, trailing 60-mo z-score	Brave & Butters (2011)	S2
43	STLFSI4	St. Louis Fed Financial Stress Index v4	FINANCIAL CONDITIONS COMPOSITE	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Kliesen & Smith (2010)	S2

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
44	SPF_BOND_M0	SPF Mean nowcast — Moody's Aaa corporate bond yield, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
45	SPF_CORECPI_M0	SPF Mean nowcast — core CPI, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
46	SPF_CPI_M0	SPF Mean nowcast — CPI, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
47	SPF_EMP_M0	SPF Mean nowcast — nonfarm payrolls, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
48	SPF_INDPROD_M0	SPF Mean nowcast — industrial production index, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
49	SPF_NGDP_M0	SPF Mean nowcast — nominal GDP, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
50	SPF_PGDP_M0	SPF Mean nowcast — GDP price index, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
51	SPF_RCBM_M0	SPF Mean nowcast — real change in private inventories, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
52	SPF_RCONSUM_M0	SPF Mean nowcast — real personal consumption, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
53	SPF_REXPORT_M0	SPF Mean nowcast — real net exports, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). *(continued)*

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
54	SPF_RFEDGOV_M0	SPF Mean nowcast — real federal government consumption & investment, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
55	SPF_RGDP_M0	SPF Mean nowcast — real GDP, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
56	SPF_RNRESIN_M0	SPF Mean nowcast — real nonresidential fixed investment, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
57	SPF_RRESINV_M0	SPF Mean nowcast — real residential fixed investment, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
58	SPF_RSLGOV_M0	SPF Mean nowcast — real state & local government consumption & investment, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
59	SPF_TBILL_M0	SPF Mean nowcast — 3-month Treasury bill rate, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
60	SPF_TBOND_M0	SPF Mean nowcast — 10-year Treasury bond yield, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
61	SPF_UNEMP_M0	SPF Mean nowcast — unemployment rate, level (T+0)	FORECASTS	philadelphiafed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed SPF (Croushore 1993)	S2
62	DEXJPUS	Japanese Yen to U.S. Dollar spot (daily, JPY/USD)	FX CURRENCY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.10 (practitioner)	S2
63	DEXUSEU	U.S. Dollars to Euro spot (daily, USD/EUR)	FX CURRENCY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.10 (practitioner)	S2

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
64	DTWEXBGS	Nominal Broad U.S. Dollar Index (daily) — DXY proxy	FX CURRENCY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.10 (practitioner)	S2
65	DTWEXEMEGS	Nominal EME U.S. Dollar Index (daily) — JPM-EM-FX proxy	FX CURRENCY	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.10 (practitioner)	S2
66	ADS_INDEX	Philly Fed Aruoba-Diebold-Scotti Business Conditions Index	GROWTH	philadelphiafed.org	level/spread, trailing 60-mo z-score	Aruoba, Diebold & Scotti (2009)	S2
67	CFNAI	Chicago Fed National Activity Index (composite)	GROWTH	chicagofed.org	level/spread, trailing 60-mo z-score	Brave, Butters & Justiniano (2019)	S2
68	GACDFS066MSFRBPH	Philadelphia Fed Manufacturing Business Outlook (general activity)	GROWTH	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Philadelphia Fed (practitioner)	S2
69	GACDISA066MSFRBNY	NY Empire State Manufacturing Survey (general activity)	GROWTH	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	NY Fed Empire State (practitioner)	S2
70	GDP	Gross Domestic Product (quarterly, nominal, SAAR billions)	GROWTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	BEA; Chen, Roll & Ross (1986)	S2
71	GDPC1	Real Gross Domestic Product (quarterly, chained 2017 \$)	GROWTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	BEA; Chen, Roll & Ross (1986)	S2
72	GDPNOW	Atlanta Fed GDPNow (real-time GDP nowcast, FRED quarterly snapshot)	GROWTH	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Atlanta Fed GDPNow (Higgins 2014)	S2
73	INDPRO	Industrial Production Index (level)	GROWTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Chen, Roll & Ross (1986)	S2, S1-eq
74	INDPRO_SURPRISE	Industrial-production surprise (realized minus SPF)	GROWTH	FRED ALFRED + Philadelphia Fed SPF	realized first-vintage minus SPF consensus nowcast; 60-mo z-score	Chen, Roll & Ross (1986) MP channel; SPF	Ch3-surprise

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
75	OECDPRINTO01GYSAMOECD	OECD Total Industry Production ex Construction (YoY growth)	GROWTH	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	OECD (practitioner)	S2
76	TCU	Capacity Utilization, Total Industry (%)	GROWTH	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Standard practitioner (FRB G.17)	S2
77	CSUSHPINSA	S&P/Case-Shiller US National Home Price Index (NSA)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Case & Shiller; S&P (practitioner)	S2
78	HOUST	Housing Starts, total privately-owned (SAAR, thousands)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	U.S. Census (practitioner)	S2
79	HSN1F	New Single-Family Home Sales (SAAR, thousands)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	U.S. Census (practitioner)	S2
80	MORTGAGE30US	30-Year Fixed Rate Mortgage Average (Freddie Mac PMMS, weekly)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Freddie Mac PMMS (practitioner)	S2
81	PERMIT	Building Permits, new private housing units (SAAR, thousands)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	U.S. Census (practitioner)	S2
82	USSTHPI	FHFA House Price Index, all-transactions (quarterly)	HOUSING	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	FHFA (practitioner)	S2
83	00XEFDEZ19M086NEST	Euro Area 19 HICP ex energy/food/alcohol/tobacco (core, index)	INFLATION	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Eurostat (practitioner)	S2
84	CORESTICKM159SFRB	Atlanta Fed Sticky-Price CPI (12-month change)	INFLATION	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Bryan & Meyer (2010)	S2
85	CPIAUCSL	Headline CPI-U, All Items (level)	INFLATION	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Chen, Roll & Ross (1986); BLS	S2, S1-eq

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
86	CPILFESL	Core CPI-U, ex Food and Energy (level)	INFLATION	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Chen, Roll & Ross (1986); BLS	S2
87	CPI_SURPRISE	CPI inflation surprise (realized minus SPF)	INFLATION	FRED ALFRED + Philadelphia Fed SPF	realized first-vintage minus SPF consensus nowcast; 60-mo z-score	Chen, Roll & Ross (1986) UI channel; SPF	Ch3-surprise
88	EXPINF10YR	Cleveland Fed 10-Year Inflation Expectations	INFLATION	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Haubrich, Pennacchi & Ritchken (2012)	S2, S1-eq
89	PCEPI	Headline PCE Price Index (level)	INFLATION	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Chen, Roll & Ross (1986); BEA	S2, S1-eq
90	PCEPILFE	Core PCE Price Index (level)	INFLATION	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Chen, Roll & Ross (1986); BEA	S2
91	PAYEMS	Nonfarm Payrolls, total (level, thousands)	LABOR MARKET	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	BLS (practitioner)	S2
92	UNRATE	U.S. unemployment rate (level)	LABOR MARKET	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	BLS (practitioner)	S2
93	UNRATE_SURPRISE	Unemployment surprise (realized minus SPF)	LABOR MARKET	FRED ALFRED + Philadelphia Fed SPF	realized first-vintage minus SPF consensus nowcast; 60-mo z-score	Boyd, Hu & Jagannathan (2005); SPF	Ch3-surprise
94	DFF	Effective Federal Funds Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Booth & Booth (1997); FRB H.15	S2
95	DFII10	10-Year TIPS Constant Maturity Real Yield (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Gürkaynak, Sack & Wright (2010)	S2
96	DGS1	1-Year Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.15 (Treasury yield)	S2, S1-bond/CP
97	DGS10	10-Year Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Welch & Goyal (2008); FRB H.15	S2, S1-eq, S1-bond/CP
98	DGS2	2-Year Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.15 (Treasury yield)	S2, S1-bond/CP
99	DGS3	3-Year Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.15 (Treasury yield)	S2, S1-bond/CP

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). (*continued*)

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
100	DGS3MO	3-Month Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Welch & Goyal (2008); FRB H.15	S2, S1-eq
101	DGS5	5-Year Treasury Constant Maturity Rate (daily)	MONETARY POLICY RATES	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	FRB H.15 (Treasury yield)	S2, S1-bond/CP
102	BBD_DISAGREE_CPI	BBD US EPU Forecaster Disagreement subindex — CPI inflation	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
103	BBD_DISAGREE_FEDSTATE	BBD US EPU Forecaster Disagreement subindex — Federal & State/Local Spending	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
104	BBD_EPU_CAT_FISCAL	BBD US EPU Categorical sub-index — Fiscal Policy (Taxes + Spending)	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
105	BBD_EPU_CAT_HEALTHCARE	BBD US EPU Categorical sub-index — Health Care	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
106	BBD_EPU_CAT_MONETARY	BBD US EPU Categorical sub-index — Monetary Policy	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
107	BBD_EPU_CAT_REGULATIONS	BBD US EPU Categorical sub-index — Regulation	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
108	BBD_EPU_CAT_TRADE	BBD US EPU Categorical sub-index — Trade Policy	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
109	BBD_EPU_COMPOSITE	BBD US Economic Policy Uncertainty Index — composite (Three-Component)	POLICY UNCERTAINTY GEOPOLITI- CAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2

Table 15: The 119 features effectively used across all pools (No., code, name, family, data source, indicator formulation, academic source, pool membership). *(continued)*

No.	Code	Factor name	Family	Data source	Formulation	Academic source	Pool
110	BBD_EPU_NEWS	BBD US EPU News-based subindex (10-newspaper headline, 1900-present)	POLICY UNCERTAINTY GEOPOLITICAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
111	BBD_EPU_TAX	BBD US EPU Tax Expirations subindex	POLICY UNCERTAINTY GEOPOLITICAL	policyuncertainty.com	level/spread, trailing 60-mo z-score	Baker, Bloom & Davis (2016)	S2
112	EMVOVERALLEMV	BBD Equity Market Volatility (EMV) Tracker — Overall (monthly FRED mirror)	POLICY UNCERTAINTY GEOPOLITICAL	fred.stlouisfed.org	Y/Y log difference, then trailing 60-mo z-score	Baker, Bloom, Davis & Kost (2019)	S2
113	IACO_GPR	Caldara-Iacoviello Geopolitical Risk Index — benchmark (10-newspaper)	POLICY UNCERTAINTY GEOPOLITICAL	matteoiacoviello.com	level/spread, trailing 60-mo z-score	Caldara & Iacoviello (2022)	S2
114	IACO_GPRA	Caldara-Iacoviello GPR Acts subindex (categories 6-8)	POLICY UNCERTAINTY GEOPOLITICAL	matteoiacoviello.com	level/spread, trailing 60-mo z-score	Caldara & Iacoviello (2022)	S2
115	IACO_GPRT	Caldara-Iacoviello GPR Threats subindex (categories 1-5)	POLICY UNCERTAINTY GEOPOLITICAL	matteoiacoviello.com	level/spread, trailing 60-mo z-score	Caldara & Iacoviello (2022)	S2
116	NYFED_ACMRNY10	NY Fed ACM 10-Year Risk-Neutral Yield (expected short-rate component)	YIELD CURVE	newyorkfed.org	level/spread, trailing 60-mo z-score	Adrian, Crump & Moench (2013)	S2
117	NYFED_ACMTTP10	NY Fed ACM 10-Year Term Premium	YIELD CURVE	newyorkfed.org	level/spread, trailing 60-mo z-score	Adrian, Crump & Moench (2013)	S2
118	RECPROUSM156N	NY Fed Recession Probability (Estrella-Mishkin)	YIELD CURVE	fred.stlouisfed.org	level/spread, trailing 60-mo z-score	Estrella & Mishkin (1998)	S2
119	TERM_SPREAD	Term spread (10Y minus 3M Treasury)	YIELD CURVE	FRED (DGS10, DGS3MO)	derived spread (level); 60-mo z-score	Fama & French (1989)	S1-eq, S1-bond/FF

A.4 References

1. Adrian, T., R. K. Crump and E. Moench (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics* 110(1): 110–138.
2. Aruoba, S. B. (2008). Data revisions are not well behaved. *Journal of Money, Credit and Banking* 40(2–3): 319–340.
3. Aruoba, S. B., F. X. Diebold and C. Scotti (2009). Real-time measurement of business conditions. *Journal of Business & Economic Statistics* 27(4): 417–427.
4. Baker, M. and J. Wurgler (2006). Investor sentiment and the cross-section of stock returns. *Journal of Finance* 61(4): 1645–1680.
5. Baker, S. R., N. Bloom and S. J. Davis (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics* 131(4): 1593–1636.
6. Baker, S. R., N. Bloom, S. J. Davis and K. J. Kost (2019). Policy news and stock market volatility. NBER Working Paper 25720.
7. Benjamini, Y. and Y. Hochberg (1995). Controlling the false discovery rate. *Journal of the Royal Statistical Society B* 57(1): 289–300.
8. Benjamini, Y. and D. Yekutieli (2001). The control of the false discovery rate under dependency. *Annals of Statistics* 29(4): 1165–1188.
9. Bernanke, B. S. (2016). The relationship between stocks and oil prices. Brookings Institution commentary, 19 February.
10. Booth, J. R. and L. C. Booth (1997). Economic factors, monetary policy, and expected returns on stocks and bonds. *Economic Review, Federal Reserve Bank of San Francisco* 1997(2): 32–42.
11. Boyd, J. H., J. Hu and R. Jagannathan (2005). The stock market’s reaction to unemployment news: why bad news is usually good for stocks. *Journal of Finance* 60(2): 649–672.
12. Brave, S. and R. A. Butters (2011). Monitoring financial stability: a financial conditions index approach. *Economic Perspectives, Federal Reserve Bank of Chicago* 35(1): 22–43.
13. Brave, S. A., R. A. Butters and A. Justiniano (2019). Forecasting economic activity with mixed frequency BVARs. *International Journal of Forecasting* 35(4): 1692–1707.
14. Bryan, M. F. and B. Meyer (2010). Are some prices in the CPI more forward looking than others? We think so. *Economic Commentary, Federal Reserve Bank of Cleveland* 2010-2.
15. Caldara, D. and M. Iacoviello (2022). Measuring geopolitical risk. *American Economic Review* 112(4): 1194–1225.
16. Campbell, J. Y. and R. J. Shiller (1988). Stock prices, earnings, and expected dividends. *Journal of Finance* 43(3): 661–676.
17. Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample. *Review of Financial Studies* 21(4): 1509–1531.
18. Chen, N.-F., R. Roll and S. A. Ross (1986). Economic forces and the stock market. *Journal of Business* 59(3): 383–403.
19. Clark, T. E. and K. D. West (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics* 138(1): 291–311.
20. Cochrane, J. H. and M. Piazzesi (2005). Bond risk premia. *American Economic Review* 95(1): 138–160.
21. Croushore, D. (1993). Introducing: the Survey of Professional Forecasters. *Business Review, Federal Reserve Bank of Philadelphia*, November/December: 3–15.
22. Croushore, D. and T. Stark (2001). A real-time data set for macroeconomists. *Journal of Econometrics* 105(1): 111–130.
23. Diebold, F. X. and M. Shin (2019). Machine learning for regression: a tutorial. In *Forecasting*.
24. Estrella, A. and F. S. Mishkin (1998). Predicting U.S. recessions: financial variables as leading indicators. *Review of Economics and Statistics* 80(1): 45–61.
25. Fama, E. F. and K. R. French (1989). Business conditions and expected returns on stocks and bonds. *Journal of Financial Economics* 25(1): 23–49.

26. Frank, I. E. and J. H. Friedman (1993). A statistical view of some chemometrics regression tools. *Technometrics* 35(2): 109–135.
27. Gürkaynak, R. S., B. Sack and J. H. Wright (2010). The TIPS yield curve and inflation compensation. *American Economic Journal: Macroeconomics* 2(1): 70–92.
28. Hakkio, C. S. and W. R. Keeton (2009). Financial stress: what is it, how can it be measured, and why does it matter? *Economic Review, Federal Reserve Bank of Kansas City* 94(2): 5–50.
29. Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics* 23(4): 365–380.
30. Haubrich, J. G., G. Pennacchi and P. Ritchken (2012). Inflation expectations, real rates, and risk premia: evidence from inflation swaps. *Review of Financial Studies* 25(5): 1588–1629.
31. Higgins, P. C. (2014). GDPNow: a model for GDP “nowcasting.” *Federal Reserve Bank of Atlanta Working Paper* 2014-7.
32. Hodrick, R. J. (1992). Dividend yields and expected stock returns: alternative procedures for inference and measurement. *Review of Financial Studies* 5(3): 357–386.
33. Hoerl, A. E. and R. W. Kennard (1970). Ridge regression: biased estimation for nonorthogonal problems. *Technometrics* 12(1): 55–67.
34. Hull, B. and X. Qiao (2017). A practitioner’s defense of return predictability. *Journal of Portfolio Management* 43(3): 60–76.
35. Kelly, B. and S. Pruitt (2013). Market expectations in the cross-section of present values. *Journal of Finance* 68(5): 1721–1756.
36. Kliesen, K. L. and D. C. Smith (2010). Measuring financial market stress. *Economic Synopses, Federal Reserve Bank of St. Louis* 2010, No. 2.
37. Lettau, M. and S. Ludvigson (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance* 56(3): 815–849.
38. Ludvigson, S. C. and S. Ng (2009). Macro factors in bond risk premia. *Review of Financial Studies* 22(12): 5027–5067.
39. Mincer, J. and V. Zarnowitz (1969). The evaluation of economic forecasts. In *Economic Forecasts and Expectations*, NBER: 3–46.
40. Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, HAC covariance matrix. *Econometrica* 55(3): 703–708.
41. Rapach, D. E., J. K. Strauss and G. Zhou (2010). Out-of-sample equity premium prediction. *Review of Financial Studies* 23(2): 821–862.
42. Stock, J. H. and M. W. Watson (2002). Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics* 20(2): 147–162.
43. Welch, I. and A. Goyal (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies* 21(4): 1455–1508.
44. Wold, H. (1975). Soft modelling by latent variables: the non-linear iterative partial least squares (NIPALS) approach. In *Perspectives in Probability and Statistics*, ed. J. Gani, Academic Press: 117–142.
45. Zou, H. and T. Hastie (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society B* 67(2): 301–320.